

Does European development have Roman roots? Evidence from the German Limes

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Abstract This paper contributes to the understanding of the long-run consequences of Roman rule on economic development. In ancient times, the area of contemporary Germany was divided into a Roman and a non-Roman part. The study uses this division to test whether the formerly Roman part of Germany are more developed than the non-Roman part. This is done using the Limes Germanicus wall as geographical discontinuity in a regression discontinuity design framework. The results indicate that economic development—as measured by luminosity—is indeed significantly and robustly larger in the formerly Roman part of Germany. The study identifies the persistence of the Roman road network until the present as an important factor causing this developmental advantage of the formerly Roman part of Germany both by fostering city growth and by allowing for a denser road network.

Keywords Roman Empire · Economic development · Germany · Boundary discontinuity · Transport infrastructure · Persistence

JEL Classification N13 · N73 · O18 · R12 · R40

1 Introduction

To what extent are contemporary levels of economic development still shaped by history? Many studies have shown that the persistent effects of deeply rooted institutional, cultural, geographical and human characteristics are central to the understanding of comparative development across countries, regions or ethnic groups and also for designing effective political measures to foster long-run prosperity (Acemoglu et al. 2001; Ashraf and Galor 2013; Bleakley and Lin 2012; Dell 2010; Giuliano and Nunn 2017). However, despite the potentially

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significant cultural heritage of the Greco-Roman civilizations, their contribution to the long-run prosperity of Europe has been largely disregarded by existing research.

This study explores the long-run effects of a Roman presence on economic development. Namely, it investigates whether there are significant differences in economic development (proxied by nighttime light intensity) between former Roman and non-Roman parts of today's Germany. Empirical identification of a positive effect of the Roman presence in Germany is based on a spatial regression discontinuity approach. In this boundary discontinuity design (BDD), the German Limes acts as two-dimensional cutoff separating treated and non-treated areas.¹

Next, I investigate whether these differences can be traced back to a particular part of the Roman legacy, namely, the persistence of their road network and its influence on the growth and location of cities. To empirically test this, I establish the persistence of the Roman road network within Germany and show that these roads in general are significantly positively related to economic development in Germany. Then I show that the road network in Roman Germany is denser and, by using a city level panel data set from 800 to 2000, that cities in the Roman part of Germany have experienced stronger growth during this period.

The road network is among those Roman legacies that is most likely to have had a persistent effect on later developments. Transport infrastructure (railways and roads) is often found to have long-lasting effects on economic development (e.g., [Berger and Enflo 2017](#); [Cogneau and Moradi 2014](#); [Holl 2004](#)) by giving an advantage to those places that are connected to a railroad, or were connected earlier. It has additionally been shown that many of the major Roman roads were also used and maintained in the centuries after the break-down of the Empire (e.g., [Glick 1979](#)). In line with this, [Bosker and Buringh \(2017\)](#) show that the probability of the existence of a city is significantly higher at locations near to Roman roads. There are three major factors explaining the persistence of the Roman road network and why it provided a particularly long-lasting development advantage to the regions previously ruled by the Romans.

First, pre-existing roads represented a cost advantage as no new roads needed to be built by the elites following the Romans. In particular, during the Middle Ages most rulers lacked the natural resources, capabilities, or money to build and maintain new road networks and, thus, largely relied on the existing ones. In later periods, the elites could use those saved resources to build additional roads. This led to a denser transport network in the Roman regions, which is clearly favorable for trade and commerce.

Second, cities founded by the Romans (e.g., Cologne, Mainz, etc.) often remained among the most important and populous metropolises in subsequent centuries. Moreover, they had a central position in the post-Roman urban networks as they were connected by the Roman roads, and were therefore easier to travel to and from making them advantageous places for trade, and giving them better market access. Thus, these have grown earlier and larger leading to a higher degree of urbanization in the Roman parts of Germany. Strongly agglomerated areas usually show more economic activity and agglomeration tends to persist (e.g., [Bleakley and Lin 2012](#); [Bosker et al. 2013](#)).

Third, due to the advantages of being located on a Roman road, new cities were more likely to occur along one ([Bosker and Buringh 2017](#))—despite the fact that Roman roads usually followed a straight line, meaning that their course was often not optimal in terms of

¹ By adopting a BDD to identify the causal effect of Roman transport infrastructure, the paper adds to a growing literature that exploits geographical or political discontinuities in space to identify causal effects of certain variables on economic outcomes like e.g., luminosity, institutional quality or political attitudes (e.g., [Becker et al. 2015](#); [Dell 2010](#); [Dell et al. 2015](#); [DehdariandGehring 2016](#); [Fontana et al. 2016](#); [Michalopoulos and Papaioannou 2014](#); [Ochsner and Roesel 2016](#); [Pinkovskiy 2017](#); [Schumann 2014](#); [Wysokinska 2015](#)).

economic transport costs, because they were constructed primarily for military rather than trade purposes (Schallmayer 2011). Hence, there is a positive feedback between the Roman road network and the forces of agglomeration and urbanization that provided cities in the Roman area with a long-lasting developmental advantage.

The former Roman border through Germany consisted of two natural river frontiers, following the banks of the Rhine and the Danube, with the Limes Germanicus wall as a joining section between them. An empirically valid boundary discontinuity design requires that the studied border is drawn in an economically unsystematic way. Therefore, this paved frontier wall is the Roman border that is most suitable for identifying the “Roman Empire effect” with a border discontinuity. On the one hand, this is because the Limes, unlike many other parts of the historical Roman border, divides a contemporary country, namely Germany. Among those parts of the Roman border separating contemporary European countries, it is the only one that is not identical to an important contemporary political border (like that between England and Scotland), a major river (the Rhine or Danube) or located in a mountain area (as is the case in Romania).² Another argument is that the course and existence of the roads in all other countries is often uncertain as they are not as well researched as in Germany. The same holds true for the history of the Limes and, in particular, for pre-Roman settlement history in Germany, which is crucial to know for the identification of the effect of the Romans on city development.

Furthermore, the formation history of the Limes suggests that its course does mostly reflect strategic and essentially military rather than economic considerations (e.g., Baatz 2000; Schallmayer 2011). This is why parts of the Limes follow straight lines, i.e. were built regardless of topographical conditions.³ Finally, focusing on the border through one country significantly reduces potential biases arising from unobserved heterogeneity due to different political institutions or culture. Hence, for the identification of a causal effect of Roman presence in general and the Roman road network in particular with a border discontinuity, the Limes wall is the most promising segment of the Roman border for this analysis.

The results indicate that, indeed, economic development is significantly higher in the historically Roman parts of Germany. I can also show that the Roman road network has largely persisted until today, that the formerly Roman parts of Germany have a denser road network and that this denser road network is associated with better economic development. In addition, I am able to show that cities in the Roman area are on average larger and that this is particularly true for cities founded by the Romans and/or connected by Roman roads. I ensure that the results are not driven by existing pre-Roman or time-invariant differences in relevant observable characteristics (e.g., Celtic settlement patterns, soil quality, elevation) and are not related to the spatial distribution of Protestants, historical migration patterns and the separation of Germany into different occupied zones in the aftermath of World War II. Additionally, I conduct a placebo exercise looking at the effect when the Limes is shifted to the north and south of its actual location and do not find significant results. Finally, I show that all results of the paper are qualitatively similar if I consider the complete Roman border

² As discussed in detail in Sect. 3, the fact that the border is identical to the Danube or the Rhine in many countries gives rise to the problems of compound treatment, e.g. due to the fact that the Rhine was also the border of e.g., the states conquered by the French during the Napoleonic period. A border that is located in a mountainous area might also not be suitable for such an analysis as no large-scale economic activity can be found in mountain regions (independent of Roman treatment). Other countries divided by the historical Roman border are located in Northern Africa. However, most of the Roman parts of North Africa are located in the Sahara, making large settlements anywhere other than the coastal area almost impossible. Moreover, the findings of Bosker et al. (2013) suggest that Roman roads played no or only a very limited role for the development of cities in North Africa and the Middle East.

³ The segment that is a straight line is also clearly visible in Figure B.1. in the Appendix.

through Germany (Appendix section B.3.1). Furthermore, in Appendix B.4.3, I show that the positive and significant effect of Roman heritage on economic development also exists if I analyze all continental European countries separated by the Roman border.

The paper is organized as follows: first, in Sect. 2, I discuss relevant existing research and its relationship to the present paper. Section 3 provides historical background on pre-Roman Germany and the history of the Limes. In Sect. 4, I describe the empirical setting and introduce the data used for the empirical analysis. Afterwards, I explain the choice of the German Limes as studied border segment. In Sect. 5, I conduct the empirical analysis, discuss relevant identification issues, and interpret the results. Section 6 concludes.

2 Related literature

The paper is most closely related to recent research evaluating systematically the extent to which legacies of the Roman Empire influenced developments in subsequent periods (e.g., [Buringh et al. 2012](#); [Dalgaard et al. 2017](#); [Michaels and Rauch 2017](#)). Yet, the few existing empirical studies on this issue have produced mixed results. [Michaels and Rauch \(2017\)](#) find that the Roman influence on city development vanished in some countries (like England) but remained persistent in others (like France) where it then actually hindered economic development, because it prevented the relocation of cities to more favorable positions on the coast. [Bosker et al. \(2013\)](#) report a significantly positive, albeit not very robust effect of being located on the hub of a Roman road in Christian parts of Europe. [Buringh et al. \(2012\)](#) conclude that the spatial distribution of the Roman market economy was different from the distribution of markets in post-Roman Europe suggesting no persistence of Roman legacies into the subsequent medieval period. Finally, [Dalgaard et al. \(2017\)](#) show that in areas around Roman roads historical population and contemporary luminosity is higher providing evidence for the importance of Roman roads (and road infrastructure in general) for long-run economic development in Roman Europe. Hence, until now it has remained unclear whether the Roman legacies persisted and had positive effects, as suggested by the historical evidence.

One reason for the ambiguous results of the existing empirical studies could be that they were almost exclusively concerned with the different impacts Roman rule had on the subsequent development of areas like France or England that were part of the Roman Empire, i.e. with the intensive margin effect of the Roman Empire. Thus, none of those studies compared the historically Roman area with the non-Roman area to investigate potentially persistent effects of the Roman Empire and, hence, an ancient legacy of Europe's later economic take-off. Finally, there is a possible interaction between the Roman transport network and agglomeration forces in the shaping of the European city system that has not been fully understood until now. Being the first paper to study the extensive margin effect of particular Roman legacies, the present study closes this gap and contributes to a better understanding of the impact of the Roman Empire on comparative development across Europe.

Furthermore, the paper is closely related to the research reporting a high amount of persistence in comparative development. This fruitful line of research has already reported the persistent economic effects of institutions, population characteristics, geographical features and cultural or colonial policies (e.g., [Acemoglu et al. 2001](#); [Ashraf and Galor 2013](#); [Bleakley and Lin 2012](#); [Giuliano and Nunn 2017](#); [Nunn 2008](#)). This study expands the knowledge about origins and consequences of historical persistence by providing evidence for the economic impact of transportation infrastructure on location and growth of cities in the very long-run.

Within the persistence literature, the paper also contributes to more specific lines of research. First, it adds to research on the persistent economic impacts of transport infras-

tructure (Berger and Enflo 2017; Cogneau and Moradi 2014; Dalgaard et al. 2017; Okoye et al. 2016; Jedwab et al. 2017). Existing studies on this matter were primarily focused on the effects of the introduction of railways in 19th century Europe and colonized countries. Thus, this study is among the first to analyze the persistence and effects of roads on development in the very long-run.

Second, the paper contributes to a better understanding of the persistent development differences between the south and the north of Germany, which have not been thoroughly studied and understood until now—at least when compared to the differences between the east and the west of Germany. According to the few existing studies on this, for example, by Jacob (2010) or Cantoni (2015), the difference between the south and the north cannot be explained by the differences between Catholics and Protestants or by the higher prevalence of Imperial cities in the south of Germany.

Finally, as the paper is among the first to find that the regional development patterns and the Great Divergence between Europe and the rest of the world actually have ancient roots, it also adds to the literature on the long-run determinants of city development in Europe (e.g., Bosker et al. 2013; Bosker and Buringh 2017; Cantoni 2015; Dittmar 2011) and on the historical origins of the “Great Divergence” (e.g., Van Zanden 2008).

3 Historical background

3.1 A brief history of the limes

As a next step, it is useful to briefly shed light on the history of the Limes with respect to its formation and decline. The formation history of the Limes suggests that it did not have a systematic course as, e.g., it did not follow borders of pre-Roman cultures (Baatz 2000; Planck and Beck 1987; Schallmayer 2011). Contemporary administrative borders of states or counties do also not follow his course. The Limes originated from the fact that, after the defeat in the Battle of the Teutoburg Forest (in 9 CE), the Romans retreated to the Rhine and the Danube. As a consequence, it was difficult for them to move troops between both separated areas. The area to the east of the Rhine, especially the Wetterau area and the high black forest, constituted significant obstacles for troop movements and acted like a block between the two Roman areas to the east of the Danube and the west of the Rhine. Hence, the Romans sought to connect their territory to the west of the Rhine with that on the east of the Danube. Subsequently, in 74/75 CE the Romans—coming from Mainz—conquered the area between the Wetterau (the south of the northernmost parts of the Limes) and the High Black Forest and built new forts there (such as, e.g., Friedberg) and in the area between the rivers Rhine and Main. At the same time, they began to build new forts in the Neckar area and the south of Bavaria (Rhaetia) by sending legions from Strasbourg eastwards and from Switzerland northwards.

After the victory against the Germanic tribe of the Chatti in northern Hesse in the year 85 CE, the Roman Emperor Domitian was able to further expand the area controlled by the Romans to the north and east. He also ordered the building of new forts and developed the Limes border line by moving troops into newly constructed camps along the Limes. He also commanded the building of new roads to connect the Rhine and the Danube provinces. In the following decades, Trajan extended the line of Limes forts first on the Main and then later along and south of the Danube (e.g., Ruffenhofen).⁴ However, in around 150 CE the

⁴ He also built the so called “Odenwald-Limes” (a more or less straight line connecting the Main area with the Neckar).

Limes was moved to the east (to its final location) in order to further shorten the connection line between the Upper German and the Rhaetian province capitals Mainz and Augsburg. As mentioned earlier, many of the final segments of the Limes wall followed a straight line and were built without regard for the varied topography of the area. One reason for this was to make the surveillance of the border as well as communication between watchtowers via smoke and light signals easier. Another reason could have been to demonstrate the superior scientific knowledge of the Romans to the barbarian Germans ([Schallmayer 2011](#); [Planck and Beck 1987](#)).

To sum up, the general course of the Limes was determined by the need to establish a safe connection between the two Roman provinces of Upper Germany and Rhaetia (or their capitals) by conquering the smallest territory necessary to achieve this—coming from the already existing Roman controlled area—without having to start another war against a Germanic tribes like the Suebi. A map showing the course of the Limes (including the early Odenwald-Limes), as well as the location, name and period of construction of the Limes forts is reported in Figure B.1 in the Appendix.

The Limes wall acted as border of the Roman Empire in Germany until the second half of the 3rd century when an agricultural crisis, continuous political power struggles and military defeats against the Sassanid weakened the strength of the Romans and led to several invasions of Germanic tribes (e.g., the Marcomanni and the Alemanni) into the Roman area. Thus, it represents the border of the largest territory in Germany that the Romans were able to retain for a long period of time. In 260 CE (at the latest in 275/276 CE) the Romans gave up their forts along the Limes and withdrew again to positions west of the Rhine, south of the Danube and east of the Iller. After this date they lost most of their control over the area to the east of the Rhine. During his more than 100 years of existence the Limes constituted not only a physical but also—although there was a significant amount of controlled trade between the non-Roman and the Roman area—an economic and cultural border between the Roman and the Germanic culture during the Roman period ([Von Schnurbein 1995](#)). Most importantly, this border also marks the boundary of the Roman Road network (according to [McCormick et al. 2013b](#)) as the Romans did not build roads outside this area. It is thus not unlikely that the differences generated during this period could have persistent effects.⁵

3.2 Important pre-Roman characteristics of Germany

It is useful to understand the historical background of the formation of the Limes to ensure that there are no important pre-Roman differences between the later Roman and non-Roman parts of Germany. To do so, this section takes a closer look at the characteristics and settlement patterns of pre-Roman Germany.

A first important question in this context is whether, from the viewpoint of the Romans, there was something unfavorable about the north of Germany that made its conquest worthless. Here, it appears that the Romans were originally set on conquering a larger part of Germany establishing the border of the Empire along the river Elbe and not the Rhine and Danube [and, in fact, the Elbe constituted the border of the Empire at least three times between

⁵ Appendix B.1.2 gives additional information on the evolution of the Roman border in Germany from the beginning of the conquest of the area in around 30 BC until the end of the Roman Empire in the 5th century. I also conducted regressions for those segments of the border that were held by the Romans until the end of the Empire. This area (roughly corresponding to the part of Germany south of the Danube and west of the Rhine) experienced Roman rule for a longer period than the area that was given up after the Limes wall was invaded. Results for these two segments are almost identical to those for the whole Limes area, but, with respect to the area south of the Danube the results indeed show a larger effect suggesting that this area might have profited from longer Roman rule (more on this in Appendix B.1.2).

12 BCE and 16 CE following the conquests of Drusus, Tiberius and Domitius Ahenobarbus (Wolters 2011)]. However, in 16 CE, among others, as a consequence of the defeat in the Battle of the Teutoburg Forest, the Romans returned to their older positions west of the Rhine and south of the Danube. Nevertheless, the successors of Tiberius repeatedly tried to reconquer parts of Germania to the east of the Rhine (Riemer 2006; Wolters 2011).

Next, powerful Germanic tribes were located both in the south of Germany (e.g., the Suebi) and the north (e.g., the Cherusci or the Langobards) of Germany showing that it was possible to maintain larger settlements in both parts of Germany and not only in the south. Even before the Roman period, the area to the west and the east of the later border was settled by Celtic tribes and in the Bronze Age there were densely settled areas in the northern Germany (e.g., in the area of the Lausitz culture). Thus, there is no reason to suppose that there were intrinsic detrimental characteristics of northern Germany that made it unattractive to the Romans (and these were likely correlated with development). Rather, it seems to be the case that there were other reasons why the cost of conquering larger parts of Germany exceeded the benefits (e.g. the failure in the Battle of the Teutoburg Forest).

Another possibility could be that the south of Germany—later conquered by the Roman Empire—was already better developed or more densely settled in the pre-Roman period. If this was the case it would mean that I wrongly attribute the discontinuity in development to the Roman presence. However, there are several historical arguments suggesting that this was not the case. First, in Appendix B.2 I discuss pre-Roman and Roman settlement patterns in Germany and conclude that there was no persistence in pre-Roman, Celtic settlement patterns (Buringh et al. 2012; Rieckhoff and Biel 2001). Moreover, there was no discontinuity in settlement density between the non-Roman and Roman area in the pre-Roman Iron Age, which is also illustrated in Fig. 1. From the settlement locations depicted there it becomes evident that the Romans and the Celts had different settlement centers.⁶

Importantly, in the period before the Romans conquered Germany, the centers of the Celtic culture shifted from the south of Germany (today's Württemberg and Bavaria) to the north west and east of the Celtic area. For uncertain reasons it was probably the case that south western Germany (the area south of the Main and south the Danube) was almost uninhabited before the Romans conquered the area.⁷

Finally, the most important technology of this era, the knowledge of how to produce iron tools and weapons, had already been widespread throughout the whole of Germany for centuries, when the Romans began their expansion (e.g., Kristiansen 2000). Thus, it seems unlikely that the Roman parts of Germany were already better developed prior to the Roman presence.

4 Data and empirical setting

4.1 Empirical setting

The unit of analysis is a grid cell of 30×30 arc seconds (0.0083 degree size).⁸ If not already available in this resolution all the data is aggregated to this grid cell size. All different

⁶ The sources for the Celtic and Roman settlements and their locations are described in Appendix B.2.

⁷ Archaeologists and pre-historians speak of the “Helvetier-” and “Boier-Einöde” (“Helvetii-” and “Boier desert”) to indicate that there are almost no findings from this period in this area (Rieckhoff and Biel 2001; Rieckhoff 2008) which was later one of the core areas of Roman Germany. And Buringh et al. (2012) note that this area of Germany was “largely devoid of settlements before the Romans arrived” (Buringh et al. 2012, p. 12).

⁸ On the equator this is equivalent to a grid cell area of 0.86 km^2 .

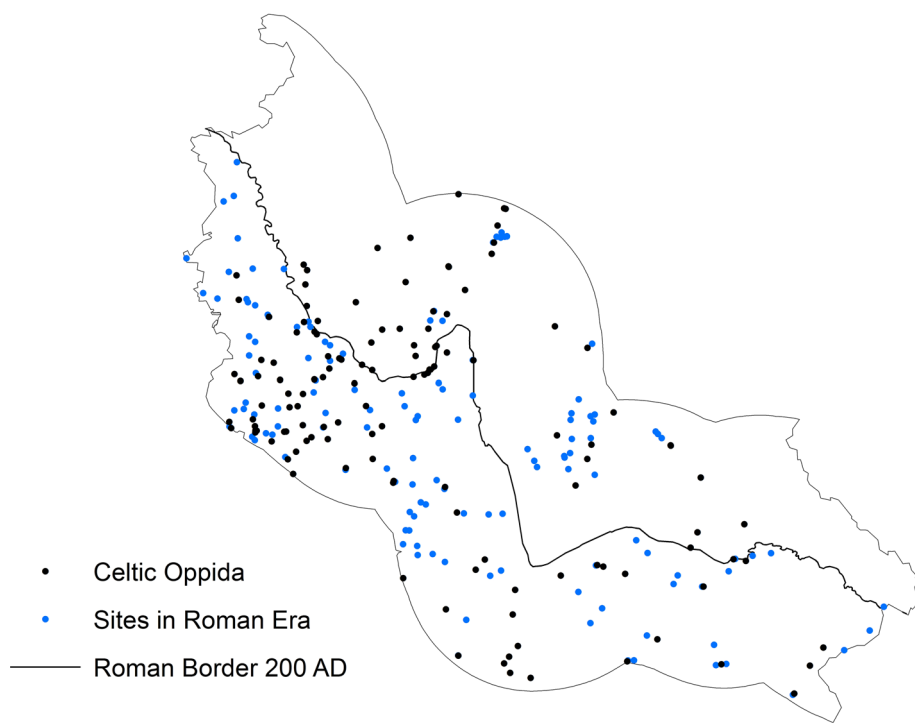


Fig. 1 Celtic settlements and Roman settlements 100 km around the Roman border in 200 CE

shapefiles are projected to use the same spatial reference (UTM WGS 1984 Zone 32N). All distances are calculated as geodesic ones. All data was obtained using ArcGIS. The borders of contemporary Germany are extracted from a shapefile of European countries provided by the Eurostat GEOSTAT database.

4.2 Data

The dependent Variable is the nighttime luminosity of a grid cell. Luminosity is measured by an integer scale ranging from 0 (unlit) to 63. Luminosity Data is available from the National Geophysical Data Center (NGDC) of the National Oceanic and Atmospheric Administration of the US. The data comes from satellite images taken for the Defense Meteorological Satellite Program (DMSP) of the US Department of Defense (the official data set is called DMSP-OLS). Here, I use the latest version of the data (4.0) and take the values of 2009. Nighttime Luminosity data is left-skewed and approximately Pareto distributed with many low values and few very large.⁹ Figure 2 shows the historical Roman border through Germany, the border of the contemporary German states, the course of the Danube, Elbe, Oder and Rhine and the distribution of luminosity in 2009. The course of the Border of the Roman Empire in 200 CE originates from a shapefile provided by Euratlas-Nüssli (Nüssli and Nüssli 2008).

⁹ The kernel density estimate of the probability distribution of luminosity is reported in Figure A.1 in the Appendix. There are also around 7% of zero values. However, these zeros very likely do not reflect areas with absolutely no economic activity but are due to some “bottom-coding” because satellites cannot distinguish very low levels of luminosity (roughly below 3) from luminosity. More on this in Appendix A.1.1.

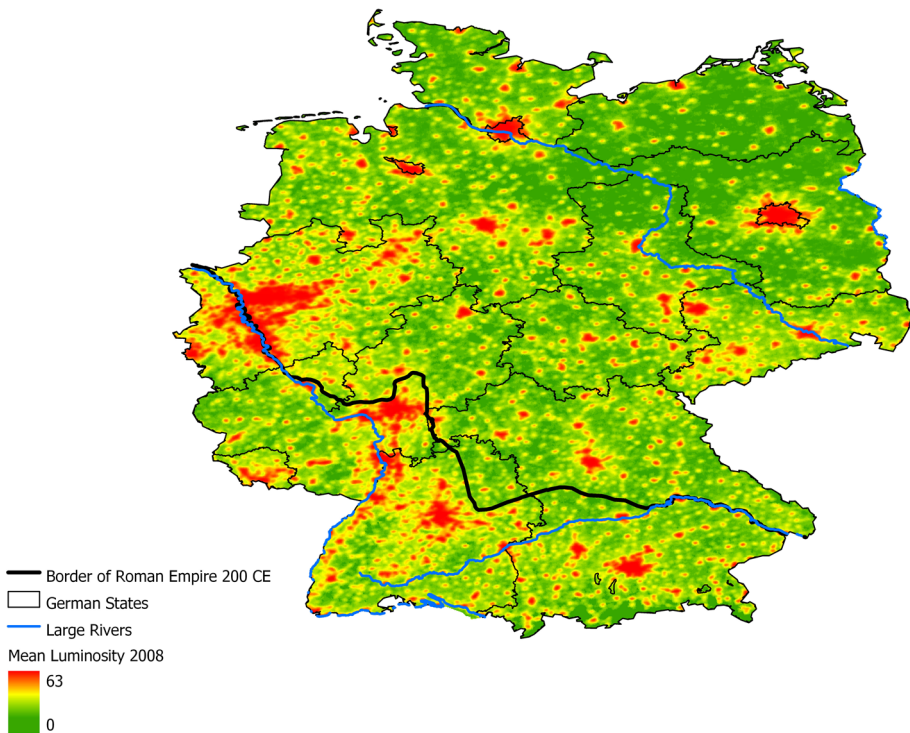


Fig. 2 Roman border, major rivers, German states and night light luminosity in 2009

Nüssli and Nüssli (2008) is a shapefile containing—among others—the border of the Roman Empire in 200 CE and is based on more than 20 historical sources, such as, e.g., the atlases of Talbert (2000) or Scarre (2000), works on Roman history like Gibbon (1983) or the Tabula Peutingeriana.¹⁰ However, I checked the validity of the drawn border by comparing it with the course of the Limes shown on the website of the “Deutsche Limeskommission” (“German Limes Committee”) (<http://www.deutsche-limeskommission.de>).¹¹

Additionally, I assembled the geographic coordinates of forts along the Limes as identified by the Limeskommission and compared whether their location corresponds with the Limes in the same way as in the map of the Limeskommission (as sometimes the forts were not directly located on the final Limes line but within a certain distance of the border). Figure A.2 in the Appendix shows the border of the Roman Empire through Germany in 200 CE and the location of Roman forts along the Limes. The map suggests that most of the locations of the forts match the course of the Limes sufficiently and in a similar way as in other sources like the map of the Limeskommission or the map of the Limes and the military sites in the Limes area in Schallmayer (2011, p. 34). Hence, the course of the Limes as drawn in the Nüssli 2008 shapefile seems to be appropriate. However, I compare the course of the

¹⁰ All sources on which the shapefile is based are available here http://www.euratlas.net/shop/bibliography/gis_200.html. A generally positive review of the collection of digital maps on which the shapefile is based is Gibbs (2011).

¹¹ A map showing the course of the Limes is available from the “Deutsche Limeskommission” (German Limes committee) here <http://www.deutsche-limeskommission.de/index.php?id=6&L=1>; accessed on September 26th, 2015.

Limes as reported by [Nüssli and Nüssli \(2008\)](#) with that depicted in the Digital Atlas of Roman and Medieval Civilizations (DARMC) ([McCormick et al. 2013a](#)) another often-cited high quality collection of historical digital maps. The comparison suggests several minor differences between both courses that should not influence the results in a decisive way. To show this explicitly, I conduct a robustness check where I estimate the BDD using the DARMC version of the Limes (Results are available in Table B.5 in the Appendix). The results are virtually identical.

From the beginnings, the Limes border line was physically visible, first as a forest aisle controlled by wooden watch towers and finally as a line of stoned towers connected by a ditch followed by an earth wall (Upper Germany) or by a stone wall (Rhaetia) ([Schallmayer 2011](#)). In consequence, the course of the Limes can be reconstructed surprisingly precisely because the Limes left physical traces like the remains of the walls and towers, surviving ditches and forest aisles as well as hills with stone rubble at the location of watch towers. Furthermore, several of the Roman forts along the Limes were reconstructed by archaeologists ([Henrich 2012, 2014](#)).¹²

The elevation data is taken from SRTM 90m data set in its newest version (4.0) and is available from the Consortium for Spatial Information (CGIAR-CSI). Data on terrain ruggedness is based on the above elevation data set by computing the grid level standard deviation of elevation. Data on agricultural suitability is computed from the data set of [Zabel et al. \(2014\)](#) and measures the suitability of a grid cell's soil for the cultivation of 16 different kinds of crops. Data on the course of major rivers (the Rhine, Danube, Elbe and Oder) are from the "WISE Large Rivers and Large Lakes" GIS map provided by the European Environment Agency.

Data on the course and coordinates of certain Roman roads is taken from the shapefile of [McCormick et al. \(2013b\)](#) who digitized the information in the "Barrington Atlas of the Greek and Roman World".¹³ Finally, the data on major roads/ highways in Germany is from the "World Roads" shapefile included in ESRI Data & Maps (ArcGIS online).

Table 1 provides a descriptive overview of the data set and also gives a first impression about the different characteristics of the Roman and non-Roman part of contemporary Germany with respect to the considered variables. In Appendix A, I describe the variables, their sources and definitions in more detail.

¹² Finally, I check whether the results change substantially when I exclude the area 5 km around the border to ensure that the results are not driven by inaccuracies of the exact course of the Limes. This check is somehow related to the idea of a "Donut RDD" ([Barreca et al. 2011](#)). Results turned out to be robust to this exclusion. Results of estimating Table 3 without the border area are reported in Table B.10 in the Appendix. To further test whether the results are driven by particular areas within the studied region that, e.g., have a very high or very low level of luminosity, I separate the studied region (i.e., the buffer area 100 km to the south and north of the Limes) into 20 equally sized 5 km distance bands. That is, the first distance band represents the area 0–5 km away from both sides of the border, the second one the area 5 to 10 km away and so on, and then re-run the BDD from Table 3, column (6) leaving out one of these distance bands in each regression. The results are shown in Appendix B.4.5, Table B.11. Overall, the coefficients remain quite stable and always significant. Hence, the results are not driven by particular areas of the sampling region.

¹³ The course, building and characteristics of Roman roads have been extensively studied by historians and archaeologists (e.g., [Laurence 1999](#)). From such works the Roman road network can be reconstructed with some certainty. Nevertheless, for some of the roads there is a scholarly disagreement about their exact course. These roads are classified in the Barrington Atlas as being uncertain and are excluded from the subsequent analysis.

Table 1 Descriptive statistics of outcomes and controls

Mean		All	S.E.	Sign.	Observations	
Roman	Non-Roman				Roman	Non-Roman
<i>Luminosity</i>						
15.676	10.811	11.978	0.035	***	177,870	483,332
<i>Agricultural suitability</i>						
52.172	37.583	41.51	0.059	***	177,366	481,673
<i>Distance to river</i>						
38.073	74.984	65	0.098	***	178,347	481,014
<i>Elevation</i>						
450.324	182.91	254.856	0.671	***	177,870	483,248
<i>Ruggedness</i>						
17.218	9.007	11.216	0.05	***	177,870	483,248
<i>Distance to highway</i>						
9.53	11.711	11.121	0.025	***	178,347	481,014

Coefficient is statistically different from zero at the ***1% level. The unit of observation is a pixel of 0.86 km² size. The standard errors reported are from a *t* test of equality of means assuming unequal variances

5 Empirical analysis

5.1 Roman rule and contemporary economic development

5.1.1 Identification issues

To be able to identify a causal effect of a “Roman legacy” on contemporary outcomes from the BDD design, the standard assumptions of the RDD have to hold (Lee and Lemieux 2010). However, in the context of a geographic discontinuity as assignment variable, there are additional challenges for identification (Dell 2010; Keele and Titiunik 2014). The standard RDD assumption is that in the absence of treatment all outcomes would vary smoothly at the border, i.e. there would be no discontinuity at the border in any outcome (Imbens and Lemieux 2008). With a geographic border as assignment variable this corresponds to the border being drawn in a non-systematic way. Testing the continuity of outcomes is possible by running a standard RD design on relevant observables. As is evident from Table 1 there are significant differences in elevation, ruggedness and agricultural suitability between the historically Roman and non-Roman parts of Germany.¹⁴ Figure C.1 in the Appendix visualizes the distribution of these three variables across Germany and also shows the course of the Roman border. A glance at these maps gives the impression that the differences between the Roman and non-Roman areas are in general not caused by the Roman border but reflect differences

¹⁴ An additional natural factor that was of considerable importance for ancient and pre-historic societies was climate. However, I do not include a measure of climate as control variable due to several reasons. First, I am not aware of any large-scale dataset on pre-medieval climate in Europe. Existing data sets start in the medieval period and have such a high spatial resolution (e.g., 5 degrees in latitude and longitude in the case of the Guiot and Corona 2010 data set) that the whole studied area would fall into one grid cell. Second, I also control for segment fixed effects, higher order polynomials of latitude and longitude and study an area with a maximum of 200 km north-south extent, I presume that climate should not vary much within this comparatively small area.

Table 2 Testing for discontinuities in covariates at the Roman border

	(1)	(2)	(3)	(4)
Buffer Area	<15 km	<10 km	<5 km	<2 km
<i>Panel A: Elevation</i>				
Roman Area	1.566	−6.639*	−3.571	−3.916
	4.012	(3.406)	(4.473)	(5.925)
Distance Polynomial	Quadratic		Linear	
Obs.	49,181	33,783	18,018	8,368
R^2	0.030	0.020	0.006	0.001
AIC	642,367	440,831	235,700	109,752
<i>Panel B: Ruggedness</i>				
Roman Area	−1.08***	−0.966***	−0.724**	−1.083**
	(0.408)	(0.243)	(0.321)	(0.430)
Distance Polynomial	4th		Linear	
Obs.	49,181	33,783	18,018	8368
R^2	0.026	0.021	0.008	0.002
AIC	381,765	262,734	140,578	65,907
<i>Panel C: Agricultural suitability</i>				
Roman Area	1.388*	1.864**	0.991	1.284
	(0.759)	(0.743)	(0.882)	(0.799)
Distance Polynomial	4th	Cubic		Linear
Obs.	49,181	33,765	18,006	8361
R^2	0.048	0.046	0.040	0.020
AIC	439,554	302,381	162,541	76,097

Robust Standard errors are reported in parentheses. Coefficient is statistically different from zero at the ***1, **5 and *10% level. The unit of observation is a pixel of 0.86 km² size. Flexible distance polynomials are applied, i.e. it is assumed that the distance polynomial in the treated area is different from that of the not treated area

between the mountainous southern part of Germany and the North German Plain.¹⁵ This impression is confirmed by statistical evidence in Table 2.¹⁶

There I test for the continuity of elevation, ruggedness and agricultural suitability at the Roman border more formally by running a classical (one-dimensional) RDD with these

¹⁵ This is probably not true for a small area in today's Hesse (approximately at the point where the border has a turning point to north–south instead of east–west orientation. This area is the “Wetterau”, which is an area with particularly high soil quality that the Romans wanted to secure for their own purposes. However, as I control for agricultural suitability this should be no concern. Furthermore, I show that the positive discontinuity in economic development also holds if I focus only on the segments of the border without the Wetterau area (see Table B.12 in the Appendix).

¹⁶ As throughout the paper, the order of the distance polynomial used in the respective columns is chosen according to the Akaike information criterion (AIC). The AIC is most commonly used to select the order of polynomials in RDD analyses (Lee and Lemieux 2010). It enables to choose among statistical models by balancing the fit of the model (measured by the variance of the residuals) and the complexity of the model (measured by the number of parameters). Its formula is $AIC = N \ln(\hat{\sigma}^2) + 2p$. Where $\hat{\sigma}^2$ is the mean squared error of the regression and p is the number of parameters in the regression model (Lee and Lemieux 2010). The model with the lowest AIC value is selected.

variables on the left-hand side.¹⁷ I estimate the BDD specification for different distance bands around the Roman border, beginning with a buffer area of less than 15 km (column (1)) and ending with a buffer of less than 2 km around the border in column (4). For all variables but ruggedness, I found no discontinuity for a buffer area of less than 5 km around the border. If I were to further narrow the estimation bandwidth to 1 km, ruggedness would also become insignificant—but given the perhaps imprecise nature of the maps of the Limes this small estimation window does not yield very reliable estimates.

Second, there could be relevant unobservable factors that cannot be tested and there is the possibility that the area later conquered by the Romans was already different from the north of Germany before the Romans. Thus, it is necessary to consider this point carefully. However, the evidence discussed in Sect. 3.2 suggests that there should be no important pre-Roman differences between the later Roman and non-Roman area of today's Germany. Finally, the formation history of the Limes suggests that its course was primarily a result of strategic goals (like, e.g. connecting the geographically separated Upper Germanic and Rhaetian province with minimal effort) and practical considerations (such as making the communication between watchtowers easy). Economic considerations or geographical characteristics seemed to be of minor importance.

Third, there should be no “compound treatment” or it should be irrelevant. Compound treatment would mean that the Roman border would completely or partly corresponds to other political/ administrative borders or geographical features that potentially matter for economic development. The historical Roman border is not identical to the borders of administrative units (states, counties) etc. of today's Germany and also not to the border of important and powerful states like Prussia or the electorate states of the Holy Roman Empire (those states whose ruler could elect the German King) like Brandenburg, Saxony or the Electoral Palatinate.¹⁸ The border is also not identical to the settlement boundaries of pre-Roman Celtic or Germanic cultures/tribes or to the borders of the Germanic and Celtic settlement areas (Hallstatt and Latene culture).¹⁹ Therefore, compound treatment should not be considered to be a main problem here.

However, the fact that the actual border of the Roman Empire through Germany—as it was after the installment of the Limes—followed the Rhine in its westernmost part and the Danube in its easternmost part (see Fig. 2) is important. As location on a river is advantageous for economic development due to several reasons, independent of Roman presence (e.g., lower transport costs, better market access etc.) it constitutes a separate treatment that cannot be distinguished from the Roman treatment when applying a BDD estimation—as distance to the Roman border and to river is identical. In principle, rivers are not expected to cause a discontinuity in economic development as areas both on the west and east of a river similarly

¹⁷ In Table C.3 I check for the continuity of soil suitability only taking into account crops that were available before the Columbian Exchange (before 2000). I do so by using the average caloric suitability index developed by Galor and Özak (2015, 2016). This measure has the additional advantage of being historically more accurate as it also assumes low input agriculture and abstracts from irrigation methods. However, it comes with the cost of a much higher spatial resolution of around 85 km². It turns out that caloric suitability for pre-Columbian Exchange crops is continuous at the Limes border for 15, 10 and 5 km bandwidths.

¹⁸ For more on this matter the reader is referred to Appendix B.3. There, Figure B.11 shows the Holy Roman Empire and its sovereign states in the year 1378, after the death of Charles IV. as well as the Roman border in 200 CE. Figure B.12 shows the borders of the states of the German Empire, in particular that of Prussia in 1871 and the Roman border in 200 CE. Finally, Figure B.18 shows the spatial distribution of Protestants and Catholics in the HRE around 1560. It also suggests that the historical Roman border did not reflect the schisms and religious upheavals of early-modern Europe.

¹⁹ The reader will find maps depicting the location of the different pre-historic cultures and the territories of Celtic settlements in Menghin (1995, p. 93 and p. 106) as well as in Kuckenburg (2000, p. 145).

benefit from its presence. However, in the particular historical setting studied here, rivers give rise to causality issues, as the Romans could have intentionally conquered the area around large rivers not only because these areas were easier to defend but also due to their favorable economic characteristics.²⁰ The decisive point here is the fact that when the Romans controlled the river this excluded the Germanic people outside the Roman Empire from profiting from the advantages of the river as they, e.g. could not use it to transport commodities. In consequence, the river might actually contribute to a larger development gap between the Roman and the non-Roman area during the Roman period; although, I would not be able to separate that part of the discontinuity caused by the river from that part caused by, e.g., the advantages of the Roman road network. Furthermore, the Rhine segment was also the border of other important historical treatments like e.g., those connected to the French invasion of Western Germany during the Napoleonic area.²¹

Thus, I decided to restrict the analysis to those parts of the border that are not identical to the course of Danube or Rhine, i.e. the Limes.²² However, in Appendix B.4.1, Tables B.1 and B.2, I present estimates showing that the results also hold if I consider the whole border including the river parts, although the resulting coefficients are somewhat smaller. Focusing on these segments of the border has the additional advantage that all the variables considered in Table 2 are now continuous for 15, 10 and 5 km buffer areas.²³

A last condition for the validity of a RDD is the absence of selective sorting, i.e. the observed units should not be able to (completely) control the assignment variable and hence their treatment status. However, it does not appear that people (or cities) could systematically choose to be located in the Roman area or not. Eventually, migration between the Roman and non-Roman parts of today's Germany was limited and controlled during the existence of the Roman Empire. This should be no valid concern here.

To further diminish heterogeneity and to account for the fact that the treatment effect might vary along different border segments (Keele and Titiunik 2014), I include border segment fixed effects in the BDD specification. Finally, I also include covariates (i.e. elevation, terrain ruggedness, agricultural suitability and distance to river) in some regressions to be sure that these factors do not cause the estimates to be biased. Results for each of the border segments separately are reported in Appendix B.4.6, Table B.12.

5.1.2 Empirical approach

The identification strategy of the paper has the following intuition: If there is a systematic difference between luminosity of the Roman and the non-Roman area, one can model luminosity as a function of distance to the Roman border and should see an upward shift in the intercept of that function at the border (i.e., at distance 0). This shift in the intercept can be estimated using a BDD. A BDD is a special case of an RDD with a two-dimensional (or multiple) forcing variable (Keele and Titiunik 2014). As there is no consensus about how to

²⁰ Buringh et al. (2012) show that settlement densities along the lower Germanic Limes, which was completely identical to the Rhine, were larger than in the regions along the landlocked parts of the German Limes. They estimated that, even when accounting for military demand, the presence of a navigable river lead to an additional 28% higher urbanization. Thus, an endogeneity problem is very likely.

²¹ For the importance of the institutional changes connected to Napoleonic rule in Western Germany on long-run development see Acemoglu et al. (2011) and Buggle (2016).

²² Descriptive statistics of the actual estimation sample for the BDD regressions can be found in the Appendix Table A.1.

²³ Results for re-estimating Table 1 for those buffer areas but without the river segments of the border are shown in Table C.1 in the Appendix.

estimate such a spatial RDD I implement all approaches applied by previous studies. First, I treat the border as a one-dimensional threshold and estimate a classical RDD with Euclidean distance to the Roman border as forcing variable. More precisely, I estimate variants of the following equation:

$$\ln(\text{Luminosity}_{s,i}) = \alpha + \beta \text{Roman}_{s,i} + f(D_i) + \gamma' \mathbf{X}_{s,i} + \delta_s + \epsilon_{s,i} \quad (1)$$

With $f(D_i)$ being a flexible function of each grid's geodesic distance to the closest border point. "Flexible" means that I allow the distance polynomial to differ in the treated and non-treated area (i.e., I interact the distance terms with the treatment variable). $\ln(\text{Luminosity}_{s,i})$ is the natural logarithm of nighttime light intensity, of each grid in border segment s in 2009.²⁴ $\text{Roman}_{s,i}$ is a dummy variable indicating whether a grid cell was located within the territory of the Roman Empire in 200 CE. $\mathbf{X}_{s,i}$ is a vector of control variables, namely distance to the closest river and grid cell i 's elevation, ruggedness and agricultural suitability. Finally, δ_s represents border segment fixed effects (where the border is split into five equally large segments).²⁵

Second, I treat the border as a two-dimensional threshold and estimate a BDD, i.e. I flexibly control for the exact geographic location of a grid cell (its longitude and latitude):

$$\ln(\text{Luminosity}_{s,i}) = \alpha + \beta \text{Roman}_{s,i} + f(x_i, y_i) + \gamma' \mathbf{X}_{s,i} + \delta_s + \epsilon_{s,i} \quad (2)$$

With $f(x_i, y_i)$ I have a flexible function of a grids' longitudinal and latitudinal coordinates (x_i and y_i).²⁶

Third, following [Seidel and Von Ehrlich \(2015\)](#) I combine both approaches and estimate an equation including both type of forcing variables.

As a robustness check, I follow [Dell \(2010\)](#) in also using non-linear distances to other geographical features or locations that are possibly relevant for economic development as forcing variable. That is, I will estimate equation (1) with $f(D_i)$ being a flexible function of each grid cell's distance to the closest major river. Results using this additional forcing variable are shown in Appendix B.4.7. The results hold and would actually imply a larger effect.

I implement the BDD in a parametric (or semi-parametric) and non-parametric way. For the parametric specifications I only consider observations less than 100km away from the border. To come closer to the theoretically ideal BDD and to show robustness of the results I also estimate it for 15, 10 and 5 km buffers around the Limes. Hence, the actual area for which the BDD is estimated is (at most) a 100km distance band around the Limes. The distribution of nighttime luminosity in this area and the Limes are reported in Fig. 3.

²⁴ Actually, to keep the zero values in the data, here and throughout the whole analysis I use $\ln(1 + \text{Luminosity}_{s,i})$ as the dependent variable. Results would be virtually identical if one would exclude the zero values from the luminosity data set (see Appendix A.1.1 for a discussion of this issue). The results would also not be affected by using the unlogarithmized version of the luminosity variable.

²⁵ As the segments that are identical to Rhine and Danube are excluded I consider only the border segments 2–4 in the BDD estimation sample. I also estimated the semi-parametric BDD using ten instead of five equally long border segments. The results of those regressions are reported in Appendix B.4.6, Table B.13. They are virtually identical to those with five border segments, if anything they are even strong with respect to the estimated size of the effect.

²⁶ I will use 2nd or 3rd order coordinates polynomial of the following form: $f(x, y) = x + y + xy + x^2 + y^2 + x^2 * y + y^2 * x + (x^3 + y^3)$.

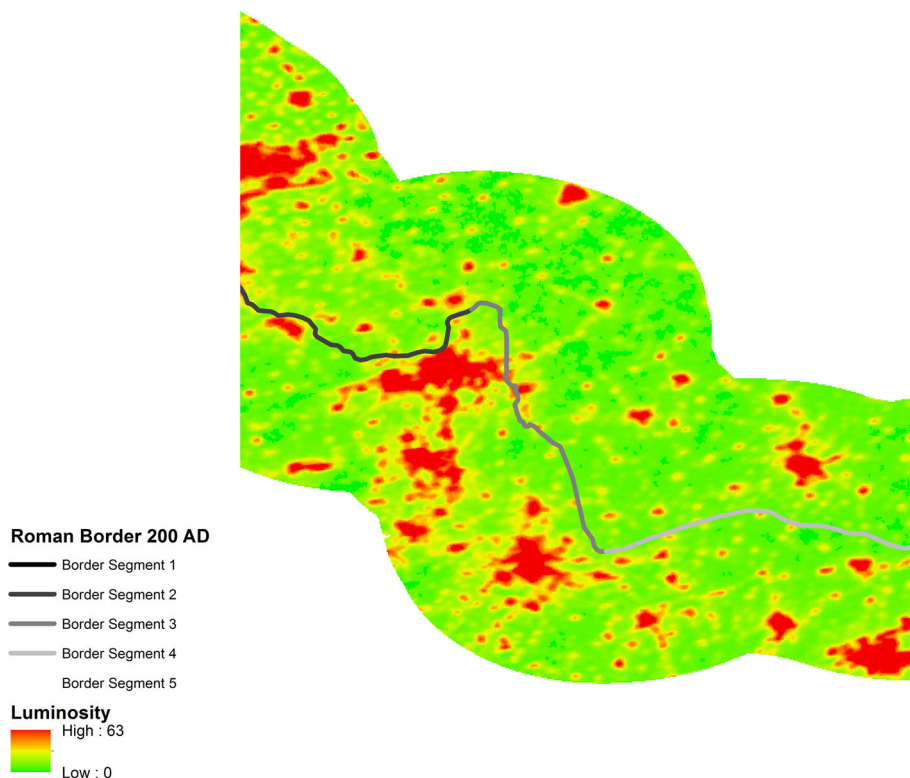


Fig. 3 Luminosity within 100 km around the Limes

5.1.3 Baseline BDD results

To get a first impression of the presence of a discontinuity in luminosity at the Roman border in Germany it is useful to plot nighttime luminosity against distance to the Roman border as is done in Fig. 4 for different bandwidths and using different methodologies. In Fig. 4a I plot nighttime luminosity against distance to border using a bandwidth (buffer area) of 100 km to the north and south of the border. The relationship between luminosity and distance to border is approximated by an 8th order polynomial chosen according to the AIC criterion. Figure 4b depicts the relationship for a 10 km bandwidth and models luminosity as a linear function of distance to border (again implied by the AIC criterion). Finally, in Fig. 4c I visualize the result of a non-parametric BDD estimation using local linear regression (LLR) and choosing the bandwidth according to the method of [Imbens and Kalyanaraman 2012](#).²⁷ All figures show a significant positive discontinuity in nighttime luminosity at the Roman border providing some initial evidence for a persistent positive effect of Roman roads on economic development.

In Table 3 I report the results of estimating non-parametric and parametric BDD specifications. In column (1) the results of the non-parametric BDD applying the LLR method are reported. The coefficient indicates that in the historically Roman area, luminosity is on

²⁷ In all figures the bins are chosen according to the IMSE-optimal evenly-spaced method using polynomial regression.

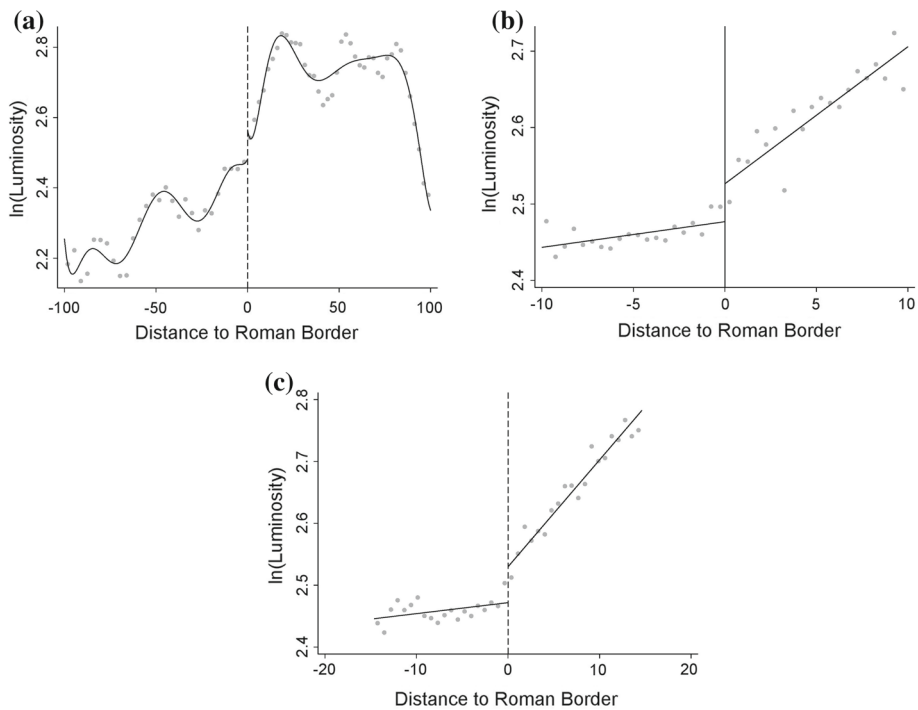


Fig. 4 Baseline BDD estimates. **a** 100 km Buffer, **b** 10 km buffer, **c** non-parametric BDD (local linear regression)

average around 5% higher than in the non-Roman area. This is virtually unchanged if I estimate the BDD using the method introduced by [Calonico et al. \(2014a\)](#) with bias-corrected robust standard errors (see [Calonico et al. 2014b](#)) correcting for too large bandwidth choices (second row of column (1)). In the case of the parametric BDD I first report the results using the coordinates polynomial, then using the distance polynomial and finally combining both in column (4). In column (5) I add border segment fixed effects and finally in column (6) I add four control variables (agricultural suitability, distance to a major river, elevation and ruggedness). The results of the parametric estimation imply an even larger effect of Roman treatment of around 10% higher luminosity in the historically Roman area.²⁸ In the case of the parametric BDD it is notable that the coefficient is much larger when only the coordinates

²⁸ Is this a large effect when compared to other historical treatments that happened long ago? To get an idea of this it is useful to compare this effect to those found by other studies on long-run impacts of treatments that happened a very long time ago. [Ashraf and Galor \(2013\)](#) assess benefits and costs of within country genetic diversity that was determined thousands of years ago by the “Out of Africa” migration process and find very large effects on contemporary development levels, especially for countries away from the optimum. They also find that around 16% of today’s variation in income per capita is explained by genetic diversity. [Putterman \(2008\)](#) finds that between one third and half of cross-country income p.c. differences in 1997 is explained by different transition years to agriculture. When I estimate a bivariate regression of the distance to Roman border variable on the natural logarithm of luminosity I find that around 6% of the variation in nighttime luminosity is explained by distance to the Roman border. In light of the results of the other studies, this seems not to be out of range (regressions available upon request). Finally, [Bosker et al. \(2013\)](#) and [Bosker and Buringh \(2017\)](#) explain the location and long-run development of cities in Europe and the MENA region. Their estimates for the effect of being located on a Roman road and a hub of Roman roads is around 0.10 and hence in a similar range to the effect estimated in this study.

Table 3 Parametric and non-parametric BDD

Dep. Var.	ln(Luminosity)					
	(1)	(2)	(3)	(4)	(5)	(6)
Method	Nonparametric RDD			Parametric RDD		
Roman Area	0.0488*** (0.014)	0.2666*** (0.006) [0.041]	0.0986*** (0.022) [0.061]	0.0988*** (0.021) [0.058]	0.109*** (0.0187) [0.055]	0.102*** (0.017) [0.047]
Roman Area (B. Corr. & Robust)	0.0535*** (0.0143)					
Order of Coordinates Polynomial		3rd		3rd	3rd	3rd
Order of Distance Polynomial			8th	8th	8th	8th
Segment Dummies	No	No	No	No	Yes	Yes
Controls	No	No	No	No	No	Yes
Obs.	29,766	181,950	181,950	181,950	181,950	181,947
Bandwidth Estimator	IK					
Bandwidth	14.618					
R ²		0.161	0.081	0.173	0.185	0.359
AIC		397,329	413,963	394,868	392,118	348,426

In columns (2)–(6) robust Standard errors are reported in parentheses and standard errors adjusted for multiway clustering along latitude and longitude are shown in brackets. Coefficient is statistically different from zero at the ***1, **5 and *10% level. The unit of observation is a pixel of 0.86 km² size. Flexible distance polynomials are applied, i.e. it is assumed that the distance polynomial in the treated area is different from that of the not treated area. Nonparametric RDD means that the coefficients in column (1) are estimated by running two separate regressions. First, a standard local linear regression (row 1) and second, a local linear regression according to the method introduced in [Calonico et al. \(2014a, b\)](#) with standard and bias-correct robust confidence intervals (row 2). Bandwidth is select according to the the selection criteria of [Imbens and Kalyanaraman \(2012\)](#). The included control variables in column (6) are agricultural suitability, ruggedness, elevation and distance to a major river

polynomial is included but not the distance to Roman border. This indicates that the distance polynomials take away much more of the treatment effect than systematic geographic patterns accounted for by the latitude and longitude polynomial.

Furthermore, standard errors clustered on latitude and longitude are reported in brackets to account most flexibly for the possibility of spatial clustering. These standard errors are estimated by applying the multiway-clustering method of [Cameron et al. \(2011\)](#). Although the standard errors are notably larger the coefficients remain significant in all but one case (column (2) without controls and the distance to border polynomial).

In Table 4 I repeat the parametric BDD estimations for smaller buffer areas of 15, 10 and 5 km around the border. I start in the upper half of Table 4 by first including the coordinates polynomial (columns (1) to (3)) and then the distance polynomial (columns (4) to (6)) together with border fixed effects. In the lower half of the table I include both distance and coordinates polynomials jointly and add control variables in the last three columns. In general, these estimations again show a significant positive effect that is in the range of the initial non-parametric result implying an effect of Roman presence of around 4–5%.

5.1.4 BDD results—robustness

While the baseline results seem to be robust to control variables, segment fixed effects, different bandwidth choices and estimation methods, there could still be sources of systematic bias that have to be explicitly addressed. First, it could be that while regional levels of development tend to be highly persistent over the long-run the results arise due to the fact that I have chosen luminosity values from 2009, especially as 2009 was the height of the global financial crisis. Second, one major shortcoming of the standard luminosity measure is that it is top-coded at the value 63. Hence it understates the luminosity of large cities and agglomerations. Third, and especially in developed countries, there is a high correlation between population density/agglomeration and economic development. Thus, the luminosity measure does capture both population concentration and economic performance. To be able to interpret the luminosity values in a more clear-cut way, it could therefore be valuable to control for the population in each grid. To address these concerns, in Table 5, I re-estimate the regressions reported in Table 3 this time using radiance-calibrated luminosity values from 2010 as dependent variable and including the natural logarithm of a grid cell's population in 2010 (column 6).²⁹ Radiance-calibrated luminosity has the advantage that it is not top-coded and hence does not suffer from a bias arising from this.³⁰

The Results in columns (1)–(5) are similar to those in Table 3 both regarding the size and significance of the effects. Eventually, the introduction of population as control variable in column (6) considerably reduces the coefficient of the Roman dummy to around 0.03.

²⁹ As the year 2010 might still be too close to the financial crisis I also re-estimate Table 3 using luminosity values from the year 2000. Results are available in Table C.2 in the Appendix and show a qualitatively similar results even suggesting a slightly larger development advantage of Roman Germany. Grid cell population originates from version 4 of the Gridded Population of the World shapefile for 2010. It can be downloaded after free registration from here: <http://beta.sedac.ciesin.columbia.edu/data/set/gpw-v4-population-count/data-download> (accessed last on October 20th, 2016). As in the case of luminosity I include $\ln(1 + Population_{s,i})$ into the regression to keep grid cells with zero population. Only 0.006% of all grid cells have a population value of zero. Hence, the results do not change when leaving them out (regressions available upon request).

³⁰ The radiance calibrated luminosity data is provided by the same source as the standard luminosity. It can be downloaded from this url: http://ngdc.noaa.gov/eog/dmsp/download_radcal.html (accessed on September 10th, 2016). However, one of the shortcomings of the radiance-calibrated luminosity series is, that it is measured with a slightly lower quality than the top-coded luminosity measures. The bivariate correlation between standard luminosity and radiance calibrated luminosity is very high (0.847).

Table 4 Semiparametric BDD estimates

Dep. Var.	ln(Luminosity)					
	(1)	(2)	(3)	(4)	(5)	(6)
Buffer Area	<15 km	<10 km	<5 km	<15 km	<10 km	<5 km
Type of Polynomial	Latitude & Longitude		Distance to Roman Border			
Order of Coordinates Polynomial	2nd		1st			
Order of Distance Polynomial						
Roman Area	0.248*** (0.008)	0.207*** (0.009)	0.115*** (0.011)	0.0472*** (0.013)	0.0476*** (0.015)	0.0299 (0.019)
Controls	No	No	No	No	No	No
Obs.	30,524	20,767	10,929	30,524	20,767	10,929
R ²	0.214	0.239	0.280	0.169	0.184	0.182
AIC	55,490	35,453	16,893	57,158	36,892	18,294
Type of Polynomial	Distance to Border & Latitude and Longitude					
Order of Coordinates Polynomial	2nd					
Order of Distance Polynomial	1st					
Roman Area	0.0543*** (0.012)	0.0552*** (0.014)	0.0328* (0.018)	−0.00109 (0.01)	0.0354*** (0.012)	0.0368*** (0.015)
Controls	No	No	No	Yes	Yes	Yes
Obs.	30,524	20,767	10,929	30,524	20,767	10,929
R ²	0.226	0.246	0.282	0.482	0.503	0.505
AIC	55,015	35,266	16,867	42,765	26,611	12,808

Robust Standard errors are reported in parentheses. Coefficient is statistically different from zero at the ***1, **5 and *10% level. The unit of observation is a pixel of 0.86 km² size. Flexible distance polynomials are applied, i.e. it is assumed that the distance polynomial in the treated area is different from that of the not treated area. All regression include border segment fixed effects. The included control variables are agricultural suitability, ruggedness, elevation and distance to a major river

However, it remains economically and statistically significant. One should keep in mind that population is also a potentially endogenous variable that reflects the impact of many Roman legacies (bad control problem) and thus it is not surprising that it lowers the estimated direct effect.

Several other important aspects are to be addressed. First, the impact pre-Roman Celtic settlements might have had on long-run development even though a descriptive look at the pre-Roman settlement patterns suggests no persistence into the Roman area (see Appendix B.2). Hence, I re-estimate the regression in Table 5 column (6) controlling for the distance of each grid cell to the closest Celtic settlement and then I additionally exclude grid cells with high luminosity values from the estimation sample. Results are shown in Table 6. In column (1) I include the distance to Celtic settlements as control variable. The coefficient of the Roman dummy remains almost unaffected by this, suggesting that pre-Roman settlements are not the cause of the estimated effects. In columns (2)–(4) I then exclude different subsamples of grid cells beginning with the 25% of cells with the largest luminosity values in column (2), the cells with above average luminosity (column (3)) and finally those cells with luminosity above the median (column (4)). Again, the results are virtually unaffected and the resulting coefficients are even slightly larger.

Second, the question is whether the results are simply due to few, very large agglomerations on the Roman side of the border (namely Frankfurt am Main, Mannheim, Munich and Stuttgart). If this is the case it is doubtful whether the estimated effects are actually due to the Roman road network, especially if those cities owe much of their current size to factors unrelated to the Romans. However, as outlined in more detail in Appendix B.4.9, of these large cities, only Mannheim has not (at least indirectly) profited from the Roman presence or is of Roman origin. Still, it could be valuable to look at what happens if these large agglomerations in the Roman area are excluded from the sample. Using the not top-coded luminosity data enables to better test this as the night light intensity in those large urban agglomerations is notably underestimated by the standard luminosity measure. Therefore, in column (5) of Table 6, I estimate the BDD from Table 5, column (6) but leaving out the grid cells located within the area (municipal borders) of those four cities. The estimated coefficient is marginally larger than that with these areas (0.0479 vs. 0.0332) and highly significant. However, to also account for the agglomerations around these cities, I exclude not only these four cities but the 35 with the highest average municipal luminosity levels in the Roman area (i.e., those of the 1% of cities with the highest average radiance calibrated luminosity that are located in the Roman part of Germany).³¹ By doing this, almost all of the smaller cities neighboring one of those four large cities (like Ludwigshafen am Rhein neighboring Mannheim or Ostfildern neighboring Stuttgart) are excluded as well as Mainz. Results are reported in column (6) of Table 6 and again show a similar coefficient that is highly significant (0.041). Thus, the results are not driven by these large agglomerations.

Third, the historically determined distribution of Protestants and Catholics within Germany could be a concern. It is well known that historically Protestant and Catholic areas

³¹ There are 3,995 municipalities within our 100km buffer area around the Limes. Average municipality level radiance calibrated luminosity is 20.481 with a standard deviation of 17.61. The 40 cities with the highest luminosity levels (1% of all municipalities) are those with values larger than 96.5357 (the maximum is 186.986). Among them, 35 are located in the Roman area and five in the non-Roman area. Municipal borders are taken from a shapefile provided by official sources (the “Bundesamt für Kartographie und Geodäsie”). It can be downloaded from here: http://www.geodatenzentrum.de/geodaten/gdz_rahmen.gdz_div?gdz_spr=deu&gdz_akt_zeile=5&gdz_anz_zeile=1&gdz_unt_zeile=13&gdz_user_id=0; accessed on December 19th, 2016.

Table 5 Parametric and non-parametric BDD with radiance calibrated luminosity and population as additional control

Dep. Var.	ln(Radiance Calibrated Luminosity) in 2010				
	(1)	(2)	(3)	(4)	(5)
Method	Nonparametric RDD		Parametric RDD		
Roman Area	0.096*** (0.162)	0.383*** (0.007)	0.130*** (0.024)	0.131*** (0.023)	0.120*** (0.021)
Roman Area (B. Corr. & Robust)	0.096*** (0.187)				0.0304** (0.015)
Order of Coordinates Polynomial		3rd		3rd	3rd
Order of Distance Polynomial			7th	7th	7th
Segment Dummies	No	No	No	No	Yes
Baseline Controls	No	No	No	No	Yes
Grid Cell Population	No	No	No	No	Yes
Obs.	28,450	181,948	181,948	181,948	181,938
Bandwidth Estimator	IK				
Bandwidth	13.921				
R^2		0.168	0.08	0.177	0.188
AIC		470,140	488,597	468,208	465,826
					0.558
					354,936

Robust standard errors are reported in parentheses. Coefficient is statistically different from zero at the ***1, **5 and *10% level. The unit of observation is a pixel of 0.86 km² size. Flexible distance polynomials are applied, i.e. it is assumed that the distance polynomial in the treated area is different from that of the not treated area. Column (4) shows the second stage estimate of a 2SLS IV regression where the distance to the closest contemporary highway is instrumented by distance to the closest Roman road (column (1) reports the first stage). In column (4) the centered- R^2 instead of R^2 is reported. The F-value of excluded IV is derived from a Sanderson-Windmeijer multivariate F test of excluded instruments (the excluded instrument is ln(Distance to Roman Road)). The included control variables are agricultural suitability, ruggedness, elevation and distance to a major river

Table 6 Parametric RDD with radiance calibrated luminosity—controlling for pre-Roman settlements and high luminosity

Dep. Var.	ln(Radiance Calibrated Luminosity) in 2010					
	(1)	(2)	(3)	(4)	(5)	(6)
Luminosity Values Excluded	Full Sample	Largest 25%	Above Mean	Largest 50%	4 Largest in Roman Area	99% Percentile of Luminosity in Roman Area
Municipalities Excluded						
Roman	0.0332** (0.015)	0.0386** (0.017)	0.0423** (0.017)	0.0482** (0.021)	0.0479*** (0.015)	0.0486*** (0.151)
Order of Coordinates Polynomial	3rd	3rd	3rd	3rd	3rd	3rd
Order of Distance Polynomial	7th	7th	7th	7th	7th	7th
Segment Dummies	Yes	Yes	Yes	Yes	Yes	Yes
Baseline Controls	Yes	Yes	Yes	Yes	Yes	Yes
Grid Cell Population	Yes	Yes	Yes	Yes	Yes	Yes
Distance to Celtic Settlement	Yes	Yes	Yes	Yes	Yes	Yes
Obs.	181,938	136,454	129,637	90,969	180,522	179,407
R ²	0.559	0.336	0.320	0.265	0.538	0.531

In columns (2)–(6) robust Standard errors are reported in parentheses. Coefficient is statistically different from zero at the ***1, **5 and *10% level. The unit of observation is a pixel of 0.86 km² size. Flexible distance polynomials are applied, i.e. it is assumed that the distance polynomial in the treated area is different from that of the not treated area. Nonparametric RDD means that the coefficients in column (1) are estimated by running two separate regressions. First, a standard local linear regression (row 1) and second, a local linear regression according to the method introduced in [Calonico et al. \(2014a, b\)](#) with standard and bias-correct robust confidence intervals (row 2). Bandwidth is select according to the the selection criteria of [Imbens and Kalyanaraman \(2012\)](#)). The included baseline control variables in column (6) are agricultural suitability, ruggedness, elevation and distance to a major river

differ with respect to many relevant economic and social outcomes (a review of the recent literature is [Becker et al. \(2016\)](#)). Therefore, it is possible that, e.g., the large cities in the Roman part of Germany actually became large and wealthy not because of the Romans but because they were Protestant. In order to rule this out, I collected information on the religious situation of the territories in the studied area in 1648 (after the peace of Westphalia) and, based on this information, identified whether a grid cell is located in a Protestant or Catholic area.³² This enables me to test whether (i) the share of Protestants is discontinuous at the Roman border and (ii) whether the results of Table 3 hold when controlling for Protestant grid cells. The former is done, in Table C.3 where I show that there is no discontinuity in the share of Protestant grid cells at the Roman border for 15, 10 and 5 km buffer areas around the Limes. The latter is tested in Table B.15, column (2). The Roman area dummy remains almost unchanged when controlling for Protestant grid cells, although Protestant grid cells show significantly higher luminosity values than Catholic ones.

Fifth, migration to Roman Germany could lead to biased estimates. While one can argue that migration is one of the channels through which the effect of Roman roads was transmitted into economic development, it is clear that migration is determined by many factors orthogonal or only indirectly related to Roman legacies. Thus, I have to check whether the results are robust when taking into account contemporary migration flows from non-Roman to Roman Germany.³³ To do so, I follow the approach introduced by [Dell \(2010\)](#) and exclude those grid cells from the sample that are located in contemporary German districts that have a regional net migration balance higher than the difference between the average regional migration balance of districts in the Roman and non-Roman parts of Germany. Based on this reduced sample I then re-estimate the regression in Table 3, column (6). The results are shown in Table B.15, column (5) in the Appendix and are virtually identical to those obtained with the full sample.³⁴

5.1.5 Placebo border

How can I make sure that the robust border effect I found is not due to a statistical coincidence? Typically, researchers conduct tests with placebo borders (shifting the border to the south or the north of the actual border) to see if they can find an effect then. I have done this by re-estimating the non-parametric BDD of Table 3 column (1) shifting the actual border 10,

³² Appendix B.4.8.2 discusses the sources on which this assignment is based and also shows maps of the spatial distribution of Protestant and Catholic territories and grid cells (Figures B.18 and B.19).

³³ Appendix B.4.8.4 gives more information e.g., provides a descriptive discussion of recent regional migration patterns in Germany.

³⁴ Another relevant concern refers to differences resulting from different allied policies in the years between 1945 and 1949, when Germany was occupied and split in four different occupation zones. As shown by recent research ([Ochsner and Roesel 2016](#); [Schumann 2014](#)) the allied countries pursued different policies in their occupation zones in post-war Germany and Austria (e.g., regarding openness for refugees) that had significant effects on the subsequent development of these areas. To rule out that my results reflect differences in the policies of the allied countries, I assign each grid cell in the data set to either the French, British or US occupation zone. In the next step, I include dummy variables for grid cells in the French and British zone to the regression specification reported in Table 3, column (6) and look at how the estimated coefficients change. Results are shown in Table B.15, column (3) in the Appendix. The border of the occupation zones are based on the shapefiles of [Schumann \(2014\)](#) and on the borders of contemporary German states. For more information the reader is referred to Appendix B.4.8.3. The estimated coefficient of the Roman area dummy is 0.115 ($p < 0.01$). Hence, the results in Table 3 are not driven by differences between the former allied occupation zones.

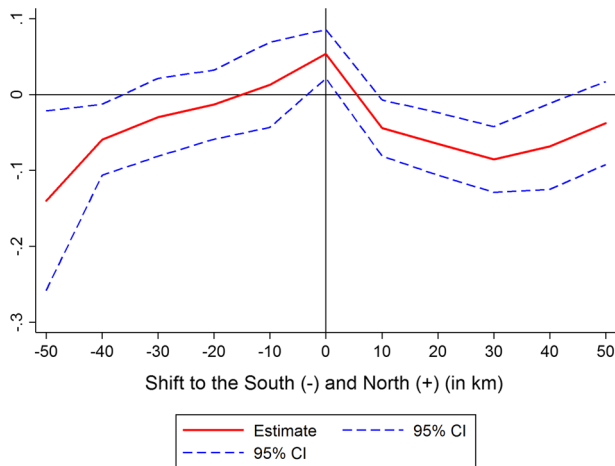


Fig. 5 Placebo test: shifting the border to the north and south

20, 30, 40 and 50 km to the north and south. The resulting coefficient estimates and 95% confidence intervals are shown in Fig. 5.

Reassuringly, I do not find a positive and significant effect for any of these shifts. However, if the border is shifted to the north I find a negative significant effects for 10, 20, 30 and 40 km. When shifting the border 50 km to the north I estimate an insignificant effect and when shifting it 50 km to the south I find a significantly negative effect.³⁵ In any case, the results show that the positive significant effect found at the historical Roman border is special and does not occur if it is drawn at placebo locations.³⁶

5.2 Channels of persistence

5.2.1 The persistence of the Roman road network

For my argument about the importance of the Roman road network, an understanding of its persistence is crucial.³⁷ In Figs. 6 and 7 I present visual evidence confirming the persistence of the Roman road network in Germany. Figure 6 shows that large parts of today's interstate highways (Autobahnen) follow the course of Roman Roads. And Fig. 7 shows that most of today's major roads (Bundesstraßen and Autobahnen) are located in the same 10 km grid as a Roman road. In fact, of the 3705 grid cells that intersect a contemporary highway, 3038 (82%) are no more than 10 km away from the location of a Roman road. And 2100 (57%) of them are located within 5 km of a Roman road. The areas for which this is not true primarily

³⁵ This is important, as the 50 km shift, ensures that only actually untreated or only treated units are included in the estimation sample, regardless of the chosen bandwidth.

³⁶ An alternative—non-parametric way—of conducting such a placebo border test is to run a Zivot–Andrews structural breakpoint test. This is done in Appendix B.4.9 and also suggests that there is a structural break in luminosity at the historically Roman border.

³⁷ Among historians, one can find different opinions about the long-run importance of Roman roads. [Bairoch \(1988\)](#) or [Lopez \(1956\)](#) for example, are skeptical about the importance of Roman roads for medieval trade. They doubt that many of the important Roman roads were maintained or represented the most cost-saving path to the trade centers.

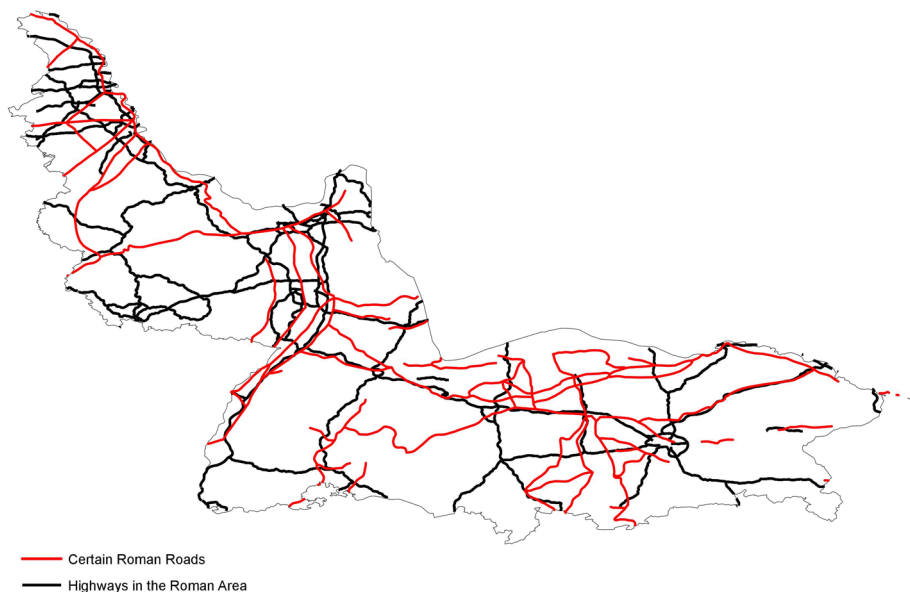


Fig. 6 Roman roads and contemporary highway

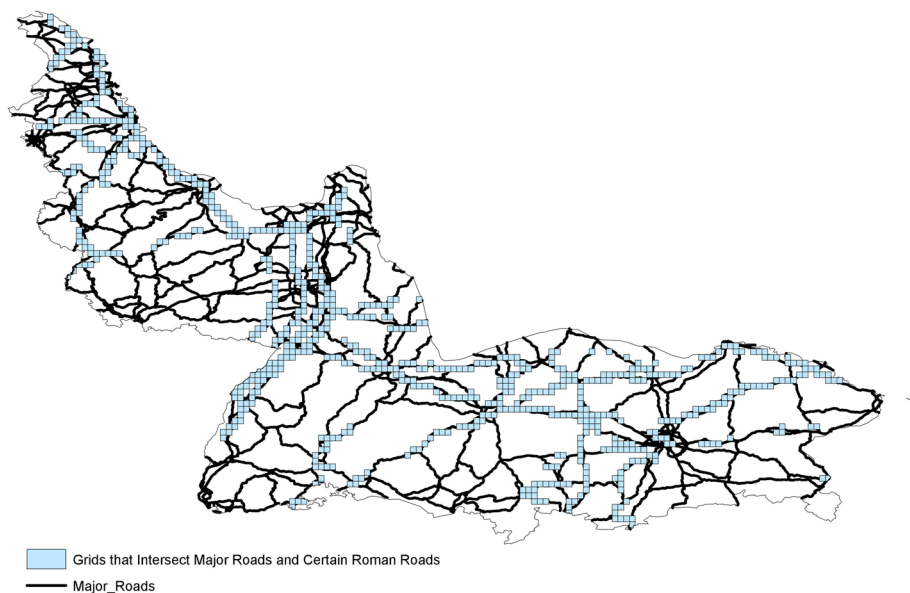


Fig. 7 Grid cells intersecting Roman roads and major contemporary roads

connect more rural areas in the south of today's Baden-Württemberg and in the south-east of Bavaria with the large agglomerations of the state capitals Stuttgart and Munich, and were also built to connect Switzerland and Austria to the major German road network.³⁸

³⁸ Furthermore, the dense road network connecting Frankfurt am Main and the Rhine-Neckar area with Saarbrücken (on the mid-west of the map) probably also follows historical Roman roads as Saarbrücken and

This persistence is likely due to the fact that many of the Roman cities and settlements remained important urban centers (Pirenne 1944; McCormick 2001). This is documented by the fact that among the southern and western German medieval trade centers many are located within the previously Roman area (such as Ulm or Ravensburg) and have a direct Roman legacy (e.g., Augsburg, Cologne, Frankfurt am Main or Regensburg). On the other hand, new cities developed along the roads connecting the Roman settlements, taking advantage of the location on a road (Bosker and Buringh 2017).³⁹ The Romans designed their roads according to military and strategic considerations, i.e. they connected forts and they aimed at taking the shortest distance to move troops. That is, why the course of Roman roads was often chosen to come as close as possible to a straight line, even if this meant accommodating steep slopes and traversing mountainous areas (e.g., Margary 1973; Lopez 1956). Therefore, the persistence of the road network is a classic case of path-dependency where the location of roads persists despite the fact that another course would be more optimal for economic purposes (e.g., transportation of goods). The idea that more cities developed in the Roman part of Germany is also supported by taking a look at the spatial distribution of city states (Free and Imperial cities) in the late medieval HRE. This is done in Figure C.4 in the Appendix. It suggests that the majority of the independent and large cities in the HRE are located in the historically Roman part of Germany and particularly in the Limes area.

Table 7 provides a more rigorous empirical test of the persistence of the Roman road network. There, I first consider (in columns 1–4) the complete historically Roman part of Germany. In columns (1)–(2) I show that there is a highly significant positive correlation between distance of a grid cell to highway or major contemporary road and its distance to a Roman road. Column (3) shows that there is a highly significant correlation between grid cells with Roman roads and those with a contemporary highway. These relationships are robust to the inclusion of border segment fixed effects and agricultural suitability, distance to a major river, elevation and ruggedness as additional controls. Then, I run a 2SLS estimation instrumenting distance to a contemporary highway by distance to a Roman road. If distance to a Roman road is a relevant instrument it is evidence for the persistence of the Roman road network. A significant coefficient in the second stage would suggest a causal effect of transport infrastructure on economic development. An instrumental variables regression also gives an idea about the indirect effect of the Roman road network on today's economic development. Column (1) also reports the coefficient from the first stage where distance to a highway is instrumented with distance to the closest Roman road. Column (4) reports the second stage results implying a very large and negative effect of distance to a highway on economic development. The F-test of excluded instruments yields a very high F-statistic documenting the strength of the instrument.⁴⁰

The results in column (5) indicate that nighttime luminosity is significantly higher in grid cells intersecting Roman roads than in grid cells that do not intersect Roman roads (when considering all of Germany as sampling area). Finally, column (6) tells us that, in general,

Footnote 38 continued

Frankfurt began as Roman settlements. However, McCormick et al. (2013b) classified these roads (or their course) as uncertain and thus I do not consider them in the analysis, leading me to underestimate the possible persistence of the road network.

³⁹ Another channel through which the persistent importance of the Roman cities could be explained is because of the surviving ecclesiastical administration in the Roman bishoprics (Cologne, Mainz, and Trier have remained important bishoprics until today).

⁴⁰ It actually also suggests that the OLS results are downwards biased as the coefficient from an OLS regression of distance to highway on luminosity with the same controls would yield a significant but notably lower coefficient of -0.263 . Regression not shown but available upon request.

Table 7 Roman legacy, transport infrastructure and luminosity

Dep. Var.	ln(Distance to Highway) (1)	ln(Distance Major Road) (2)	Highway Grid (3)	ln(Luminosity) (4)	ln(Luminosity) (5)	ln(Luminosity) (6)	ln(Distance to Highway) (7)
Sampling Region	Roman Germany			Germany			10 km Buffer
Method	OLS			2SLS	OLS		
Order of Coordinates Polynomial							
Order of Distance Border Polynomial							
ln(Distance to Roman Road)	0.116*** (0.002)	0.124*** (0.002)					2nd
Roman Road Grid			0.0139*** (0.003)		0.5661*** (0.01)		1st
ln(Distance to Highway)				−0.765*** (0.014)		−0.415*** (0.001)	
Roman Area							−0.0923*** (0.019)
Segment Dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Obs.	177,366	177,366	177,366	177,366	658,976	658,976	20,767
R ² /Centered-R ²	0.193	0.162	0.016	0.123	0.095	0.244	0.368
F-value of excluded IV				248,079			
AIC							45,985.32

In columns (2)–(6) robust Standard errors are reported in parentheses. Coefficient is statistically different from zero at the ***1, **5 and *10% level. The unit of observation is a pixel of 0.86 km² size. Flexible distance polynomials are applied, i.e. it is assumed that the distance polynomial in the treated area is different from that of the not treated area. The included baseline controls in are agricultural suitability, ruggedness, elevation, and distance to a major river

distance to a highway is significantly negatively associated with economic development.⁴¹ Figure 7 visualizes these relationships. Looking at this map it becomes evident that the centers of economic activity (corresponding to the largest agglomerations/ cities) are all connected by both highways and Roman roads. This suggests that the most important centers of economic activity today were already connected with roads during the Roman era.

5.2.2 Roman legacy and a denser road network

Now I test the first of the proposed transmission channels, namely that the persistence of the Roman road network allowed for a denser road network. To do so, I re-run the parametric BDD specification used in the lower half of Table 5 column (5), i.e. I include both the distance to the Roman border as well as the coordinates polynomial and I only consider the area 10km around the historical Roman border. The result is depicted in Table 7, column (7) and suggests that, indeed, there is a significant negative discontinuity in distance to a highway at the historical Roman border.⁴² This is suggestive evidence for the idea that the persistence of the Roman road network allowed for a denser transportation network.

5.2.3 Roman legacy and long-run city development

The second channel I consider to be responsible for the persistent effects of the Roman road network on contemporary economic prosperity is its influence on city growth and location. After the decline of the Roman Empire most of the cities/ settlements of the Romans remained important urban centers, e.g. due to the fact that almost all of them were connected by Roman roads (e.g., [Hohenberg and Lees 1995](#); [Planitz 1966](#)). Therefore, those cities were easier to reach, giving them the advantage of a better market access and making them centers of trade and commerce. These advantages allowed them to grow earlier and faster than the non-Roman cities. This in turn led them to become larger and additionally resulted in a higher degree of agglomeration and urbanization in general. This was because over time new cities were founded along the existing roads that took advantage of the location on a road and managed to become notable centers of trade. The persistence of the urban Roman network is also an additional factor explaining the persistence of the Roman road network as it is clear that the important urban centers are always connected by major roads and if these centers remain important, then the roads connecting them will be maintained and persist.

To test the significance of this channel I create a city-level panel data set including the population of cities 100 km to the south and the north of the Limes in the centuries between 800 and 2000 (there is no city population data in 1100). The city population data for each one-hundred years between 800 and 1800 originates from [Bairoch et al. \(1984\)](#) and for the years 1900 and 2000 I took the values from the Clio-infra database on urban settlements.⁴³ For the studied area these sources provide city populations for 54 cities (36 on the Roman side of the Limes and 18 on the non-Roman side). Altogether the data set consists of 648 city-year pairs. As city population figures are not available for each of the city-year pairs the actual

⁴¹ This would also work with a dummy variable indicating grids that intersect a highway. Regression not shown but available upon request.

⁴² This result also holds if one were to control for luminosity to account for the fact that the higher road density could also be the result of higher economic development that in turn could have been the result of higher levels of urbanization and agglomeration caused by Roman legacies. The inclusion of luminosity would reduce the coefficient to -0.0544 which would still be significant at the 1% level.

⁴³ The data can be downloaded here: <http://www.cgeh.nl/sites/default/files/def%20europe.xls>; accessed on July, 10th, 2015.

number of city-year pairs on which I conduct the empirical analysis is 289. I supplement the city population data with the coordinates of the cities and the same variables as used in the previous grid level analysis. That is, I include the elevation at a city's coordinates, the standard deviation of elevation (ruggedness) and agricultural suitability in an area 5 km around the city, as well as a city's distance to the closest major river and the Roman road. Furthermore, I collected information on Roman cities/ settlements and whether these were located on a Roman road.⁴⁴ I also gathered information on whether a city was located near to (i.e., not more than 10 km away from) a main Celtic settlements (oppida, hillforts, princely residences) and calculated the distance to the most important princely residence of the Celts, i.e. the Heuneburg. These variables can account for potential existing influences of Celtic culture and settlement patterns. Finally, I include the distance of each city to the closest Roman market or mine (according to DARMC) as controls to test for the presence of additional transmission channels, e.g. through the persistence of economic activities.⁴⁵ A descriptive overview of this data set is given in the Appendix, Table A.2.⁴⁶

To empirically test the persistent impact of the Romans on city development I estimate the following regression specification:

$$\ln(Population_{isc}) = \alpha + \beta Roman_{si} + \gamma' \mathbf{X}_{si} + \delta_s + \lambda_c + \epsilon_{isc} \quad (3)$$

where $\ln(Population_{isc})$ is the natural logarithm of the population of city i in border segment s in year c with $c = 800, 900, \dots, 2000$. $Roman_{si}$ is one of five measures of Roman treatment of city i in border segment s . $\mathbf{X}_{si}$ is a vector of control variables including agricultural suitability, ruggedness, elevation, a Celtic settlement dummy and distance to a major river and the Heuneburg as well as to the closest Roman market and mine. Finally, δ_s are border segment fixed effects, λ_c are year fixed effects and ϵ_{isc} is the error term.⁴⁷ Equation (3) is estimated using OLS with standard errors clustered on city level. Results of the estimations are reported in Table 8. In column (1) I regress city population on a dummy for cities located in the historically Roman area. I find a large and positive effect indicating that cities on the Roman side of the border are on average around 50% larger. If I limit myself to the cross-section of city population in 2000, the estimated effect is even larger. In column (3) I include a dummy variable for cities actually founded by the Romans, i.e. cities that developed from a Roman settlement (like e.g., Mainz or Trier) and find a comparable positive effect. I uncover a smaller, yet still economically and statistically significant effect if I limit myself only to cities located on a Roman road (column (4)).⁴⁸ However, the most direct test of my argument is to look at cities founded by the Romans that were located on a Roman road. If I include a

⁴⁴ Information about Roman settlements is taken from the shapefile "Europe in 200 AD" provided by EurAtlas Nüssli and Nüssli (2008).

⁴⁵ From the findings of the long-run city development literature one can argue that other factors, like e.g., the ecclesiastical or commercial importance of a city should also be included here. However, it is likely that these variables, e.g. a dummy for residences of archbishops or medieval trade centers, are endogenous to the Roman treatment and hence would be bad controls.

⁴⁶ Figure C.3 in the Appendix shows the locations of the cities (cities on the Roman side of the border in red and cities on the non-Roman side of the border in blue), their size in 1800 (Figure C.3a) and 2000 (Figure C.3b) indicated by the size of the dots, as well as the Roman road network. The visual impression suggests that cities on the Roman side of the border seem to be larger on average than their counterparts in the non-Roman area.

⁴⁷ In Appendix B.4.10 (Table B.17) I also include a dummy for cities becoming Protestant in the 16th century as additional control to rule out that north–south differences in religion drive the results. It turns out that Protestantism is always insignificant and that the coefficients are almost identical.

⁴⁸ I code a city as being located on a Roman road if it is located within a 5 km buffer around the road.

Table 8 Roman legacy and city development

Dep. Var.	ln(City Population)					
	(1)	(2)	(3)	(4)	(5)	(6)
Sample	All Obs.	2000 only		All Obs.		Roman Area
Roman	0.471 * (0.239)	0.873** (0.353)				
Roman City			0.597*** (0.194)			0.465** (0.202)
City on Roman Road				0.503*** (0.172)		0.318 (0.228)
Roman City on Roman Road					0.628*** (0.192)	
Distance to Roman Road						−0.0232 (0.014)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Obs.	289	54	289	289	289	200
Adj. R ²	0.681	0.576	0.696	0.69	0.7	0.732

Standard errors clustered on city level are reported in parentheses. Coefficient is statistically different from zero at the ***1, **5 and *10% level. The unit of observation is a pixel of 0.86 km² size. The included control variables are agricultural suitability, ruggedness, Celtic settlements elevation, distance to a major river, the Heuneburg and the closest Roman market and mine as well as segment and century fixed effects

dummy variable identifying those cities in the regression, the estimates (column (5)) again suggest that those cities were on average around 60% larger than the other cities.⁴⁹

To directly test the hypothesis that the advantage of Roman cities is to a large extent due to their location on a Roman road I limit the analysis to the Roman area and show that within the Roman area, Roman cities had a growth advantage compared to non-Roman cities (column (6)) and that this advantage almost disappears when I additionally include the distance to the next Roman road (column (7)).

All in all, the estimates in Table 8 indicate that city growth was larger in the formerly Roman area of Germany and that this higher city growth probably resulted from the amenities of the Roman road network. Celtic settlements, the distances to the Celtic princely residence Heuneburg and the distance to the closest Roman market or mine do not have any significant influence, confirming both the proposed transmission channel as well as the notion of non-persistence of Celtic settlement patterns. Compared to the previous findings of Bosker et al. (2013) my empirical results suggest a more robust and larger effect of Roman roads on city development than they found in their, larger European sample. Even more, their observation period ends in 1800, while I could show that the effect survived the fundamental changes connected to the Industrial Revolution and is visible even today.

Similar to the BDD analysis, two major concerns with those results remain. First, they could be driven by only a few, very large agglomerations (Frankfurt am Main, Mannheim, Munich and Stuttgart). On the one hand, I have already accounted for the fact that not all cities in the Roman area have a direct Roman legacy (e.g., are founded by the Romans or located on a Roman road) and still find a positive and significant effect. On the other hand, to ensure that the results do not depend on these few outliers, I re-run the estimates in Table 8 (and for the area around the whole German border in the Appendix) excluding those cities and Nuremberg (as largest agglomeration on the non-Roman side) from the estimation sample. Results are reported in Appendix B.4.11 and turn out to be similar. Hence, the effect is not driven by them but does also hold for medium and smaller sized cities on both side of the border.

Second, historical migration to the cities in the Roman area could drive the results. To address this I apply the same procedure as in Sect. 5.1.4 for the grid cell data set. As for past centuries no official regional migration statistics are available, I rely on historical data on the migration of famous people as proxy for overall migration to a city (Croix and Lican-dro 2015; Schich et al. 2014). I obtain information on this from the Schich et al. (2014)) data set on important life dates (like location and year of birth and death) of thousands of famous people from all over the world beginning in 800 and ending in 2000. To be precise, I count the number of famous people that died in a respective city and century but were not born there—and hence had migrated to the city at some point in time.⁵⁰ Using this information I calculate the difference between the average number of famous people migrating to a city in the non-Roman versus the Roman area (which is 3) and exclude city-century pairs having a higher number of migrating famous people. Using this reduced sample I re-estimate Table 8. The results are shown in Appendix Table B.16. While the Roman area dummy becomes insignificant now, Roman cities and cities on Roman roads (or

⁴⁹ Again, one can ask whether these large effects are reasonable. While it is right that 60% is a very large effect, it is not out of the range of effects found by previous studies on long-run city development. Bosker et al. (2013) for example, estimate that being a capital city increases population on average by around 90%. They also estimate that residence cities of archbishops or of cities located on the coast are on average around 40% larger, roughly corresponding to the size advantage I estimated for cities in the Roman part of Germany.

⁵⁰ More on these data and the construction of the famous people migration index can be found in Appendix B.4.8.4.



Fig. 8 Temporal evolution of the effect of being a Roman city on economic development 1000–2000

both) still show significantly higher population than their non-Roman counterparts and the estimated coefficient sizes are qualitatively similar. Thus, while migration seems to play a role in explaining the growth advantage of the Roman area in general it is not able to explain away the development advantage of cities that directly profited from Roman presence or legacies.

Finally, it is worthwhile to look at the long-run temporal evolution of the effect of Roman legacies on city development. In order to do so, I interact the Roman city dummy with the century dummies and repeat the regression in Table 8 column (4) including these interaction terms showing the effect of being a city in the historical Roman territory on city development for each of the centuries.⁵¹ The resulting coefficients of the interaction terms and the respective 90% confidence interval are shown in Fig. 8. Although, I included all interactions into the regression on which Fig. 8 is based I only show the coefficients for the year 1200 onward. The reason for this is that before 1200 city population figures exist only for a few important cities leading to unreliably high coefficients before 1200. The pattern in Fig. 8 suggests that the effect of the Roman road network on city development is significant—yet slightly declining—over most of the period between 1200 and 2000. The exceptions are 1700 and 1800, which can probably be explained by the distortions caused by the Thirty Years' War. Thus, the effect detected in the panel estimates is not caused by a particular period but seems to be persistent over most of the previous 800 years.⁵²

⁵¹ I do not include the Celtic settlement dummy, the distance to Heuneburg and the distance to the closest Roman market and mine variables to this regression, as they were always insignificant in Table 8—and would be insignificant in this regression.

⁵² Another way to look at the temporal evolution of the Roman effect is to run separate cross-sections for each century. This has the advantage that one can directly interpret the estimated coefficients as the effect of Roman legacies in the respective century. However, it comes with the cost of a considerably reduced sample size making the estimates more imprecise. Nevertheless, I estimate separate cross-sections of the regression in Table 8 column (1) for the whole sample of cities (including the river areas) and for the 15th century, in order to have enough observations. Results are shown in Appendix B.4.12, Table B.20. I find significant and positive effects beginning in the 16th century with a notable decrease in the coefficient size in the 18th century. Hence, the results of the separate cross-sections do in principle confirm the pattern observed when interacting the Roman dummy with the century fixed effects.

6 Concluding remarks

The present study has shown that the Roman Limes border wall across contemporary Germany constitutes a positive discontinuity in economic development. Those parts of contemporary Germany that were once part of the Roman Empire show higher economic development than the non-Roman parts. I was also able to show that this positive and long-lasting effect of Roman presence in Germany is likely due to the persistence of the Roman road network and its interaction with agglomeration forces. This persistence meant that settlements in the former Roman Empire have had developmental advantages in several ways as it has allowed for a denser transportation network and a faster city growth resulting in higher levels of urbanization, agglomeration and economic activity.

Thus, the present study is the first to explicitly documenting the positive long-run effect of the Roman road network on economic development at the extensive margin. It is also among the first studies documenting very-long lasting positive effects of transport infrastructure on the development of cities and regions. The cities in the historically Roman area grew very large and remained more prosperous than their non-Roman counterparts until today. Therefore, it was the cities of formerly Roman Germany (and probably also Roman Europe) that played a key role for the innovations and developments that later laid the ground for the Great Divergence between Europe and the rest of the world.

Still, the long-run economic effects of the legacies of ancient cultures have not been studied in much depth. Therefore, exploring the various potentially important legacies of the ancient civilizations, especially of the Romans but also of the Greeks, will be an important task of future research projects.

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References

- Acemoglu, D., Cantoni, D., Johnson, S., & Robinson, J. A. (2011). The consequences of radical reform: The french revolution. *American Economic Review*, 101, 3286–3307.
- Acemoglu, D., Johnson, S., & Robinson, J. A. (2001). The colonial origins of comparative development: An empirical investigation. *American Economic Review*, 91, 1369–1401.
- Ashraf, Q., & Galor, O. (2013). The 'Out of Africa' hypothesis, human genetic diversity, and comparative economic development. *American Economic Review*, 103, 1–46.
- Baatz, D. (2000). Der römische Limes. Archäologische Ausflüge zwischen Rhein und Donau. Berlin: Gebr. Mann Verlag.
- Bairoch, P. (1988). *Cities and economic development. From the dawn of history to the present*. Chicago, IL: University of Chicago Press.
- Bairoch, P., Batou, J., & Chevre, P. (1984). *The population of European cities from 800 to 1850*. Geneva: Librairie Droz.
- Barreca, A. I., Guldi, M., Lindo, J. M., & Waddell, G. R. (2011). Saving babies? Revisiting the effect of very low birth weight classification. *Quarterly Journal of Economics*, 126, 2117–2123.
- Becker, S. O., Boeckh, K., Hainz, C., & Woessmann, L. (2015). The empire is dead, long live the empire! Long-run persistence of trust and corruption in the bureaucracy. *Economic Journal*, 126, 40–74.
- Becker, S. O., Pfaff, S., & Rubin, J. (2016). Causes and consequences of the protestant reformation. *Explorations in Economic History*, 62, 1–25.

- Berger, T., & Enflo, K. (2017). Locomotives of local growth: The short- and long-term impact of railroads in Sweden. *Journal of Urban Economics*, 98, 124–138.
- Bleakley, H., & Lin, J. (2012). Portage and path dependence. *Quarterly Journal of Economics*, 127, 587–644.
- Bosker, M., & Buringh, E. (2017). City seeds: Geography and the origins of the European city system. *Journal of Urban Economics*, 98, 139–157.
- Bosker, M., Buringh, E., & Van Zanden, J. L. (2013). From Baghdad to London. Unraveling urban development in Europe, the Middle East, and North Africa, 800–1800. *Review of Economics and Statistics*, 95, 1418–1437.
- Buggle, J. C. (2016). Law and social capital: Evidence from the code napoleon. *European Economic Review*, 87, 148–175.
- Buringh, E., van Zanden, J. L., & Bosker, M. (2012). *Soldiers and booze: The rise and decline of a Roman market economy in north-western Europe*. CGEH working paper no. 32.
- Calonico, S., Cattaneo, M. D., & Titiunik, R. (2014a). Robust nonparametric confidence intervals for regression-discontinuity designs. *Econometrica*, 82, 2295–2326.
- Calonico, S., Cattaneo, M. D., & Titiunik, R. (2014b). Robust data-driven Inference in the regression-discontinuity design. *The Stata Journal*, 13, 1–36.
- Cameron, A. C., Gelbach, J. B., & Miller, D. L. (2011). Robust inference with multiway clustering. *Journal of Business & Economic Statistics*, 29, 238–249.
- Cantoni, D. (2015). The economic effects of the protestant reformation: Testing the Weber hypothesis in the German Lands. *Journal of the European Economic Association*, 13, 561–598.
- Cogneau, D., & Moradi, A. (2014). Borders that divide: Education and religion in Ghana and Togo since colonial times. *Journal of Economic History*, 74, 694–729.
- Dalgaard, C.-J., Kaarsen, N., Olsson, O., & Selaya, P. (2017). Roman roads to prosperity: The long- run impact of infrastructure on economic activity. *Presentation at London School of Economics*. url: https://dl.dropboxusercontent.com/u/17139768/roman_slides.pdf.
- de la Croix, D., & Licandro, O. (2015). The longevity of famous people from Hammurabi to Einstein. *Journal of Economic Growth*, 20, 263–303.
- Dehdari, S., & Gehring, K. (2016). *Building national identity? The causal effect of homogenization policies in Alsac-Lorraine*. CIS working paper no. 89.
- Dell, M. (2010). The persistent effects of Peru's mining mita. *Econometrica*, 78, 1863–1903.
- Dell, M., Lane, N., & Querubin, P. (2015). *State capacity, local governance and economic development in Vietnam*. Mimeo.
- Dittmar, J. E. (2011). Information technology and economic change. The impact of the printing press. *The Quarterly Journal of Economics*, 126, 1133–1172.
- Fontana, N., Nannicini, T., & Tabellini, G. (2016). *Historical roots of political extremism. The effects of Nazi occupation of Italy*. CEPR discussion paper, no. 11758.
- Galor, O., & Özak, Ö (2015). *Land productivity and economic development: Caloric suitability vs. agricultural suitability*. Brown University working paper.
- Galor, O., & Özak, Ö. (2016). The agricultural origins of time preference. *American Economic Review*, 106, 3064–3103.
- Gibbon, E. (Ed.). (1983). *Histoire du d'eclin et de la chute de l'Empire Romain*. Paris: Robert Laffont.
- Gibbs, P. (2011). Marc-Antoine Nüssli & Christos Nüssli. Euratlas Periodis Expert: Periodical Historical Atlas of Europe 1–2000 (digital cartography program). *Antiquity*, 85, 1101–1102.
- Giuliano, P., & Nunn, N. (2017). *Understanding cultural persistence and change*. Mimeo.
- Glick, T. F. (1979). *Islamic and christian Spain in the early Middle Ages*. Princeton, NJ: Princeton University Press.
- Guiot, J., & Corona, C. (2010). Growing season temperatures in Europe and climate forcings over the past 1400 years. *PLoS ONE*, 5(4), 1–15.
- Henrich, P., (Ed.) (2012). *Der Limes vom Niederrhein bis an die Donau*. Vol. 6. Beitrage zum Welterbe Limes. Stuttgart: Konrad Theiss Verlag.
- Henrich, P., (Ed.) (2014). *Der Limes in Raetien, Ober- und Niedergermanien vom 1. bis 4. Jahrhundert*. Vol. 8. Beiträge zum Welterbe Limes. Darmstadt: Konrad Theiss Verlag.
- Hohenberg, Paul M., & Lees, L. H. (1995). *The making of urban Europe, 1000–1994*. Cambridge, MA: Harvard University Press.
- Holl, A. (2004). Manufacturing location and impacts of road transport infrastructure: Empirical evidence from Spain. *Regional Science and Urban Economics*, 34, 341–363.
- Imbens, G. W., & Lemieux, T. (2008). Regression discontinuity designs: A guide to practice. *Journal of Econometrics*, 142, 615–635.
- Imbens, G., & Kalyanaraman, K. (2012). Optimal bandwidth choice for the regression discontinuity estimator. *The Review of Economic Studies*, 79, 933–959.

- Jacob, M. (2010). *Long-term persistence: The free and imperial city experience in Germany*. Mimeo.
- Jedwab, R., Kerby, E., & Moradi, A. (2017). History, path dependence and development: Evidence from colonial railroads, settlers and cities in Kenya. *Economic Journal*. doi:[10.1111/econj.12347](https://doi.org/10.1111/econj.12347).
- Keele, L., & Titiunik, R. (2014). Geographic boundaries as regression discontinuities. *Political Analysis*, 22, 814–863.
- Kristiansen, K. (2000). *Europe before history*. Cambridge: Cambridge University Press.
- Kuckenburg, M. (2000). *Vom Steinzeitlager zur Keltenstadt. Siedlungen der Vorgeschichte in Deutschland*. Stuttgart: Konrad Theiss Verlag.
- Laurence, R. (1999). *The roads of Roman Italy: Mobility and cultural change*. London: Routledge.
- Lee, D. S., & Lemieux, T. (2010). Regression discontinuity designs in economics. *Journal of Economic Literature*, 48, 281–355.
- Lopez, R. S. (1956). The evolution of land transport in the Middle Ages. *Past and Present*, 9, 17–29.
- Margary, I. D. (1973). *Roman roads in Britain*. London: John Baker Publishing.
- McCormick, M. (2001). *Origins of the European economy. Communication and commerce AD 300–900*. Cambridge: Cambridge University Press.
- McCormick, M., Huang, G., Zambotti, G., & Lavash, J., (Eds.). (2013a). *Digital atlas of Roman and medieval civilizations*. Cambridge, MA: DARMC, Center for Geographic Analysis, Harvard University.
- McCormick, M., Huang, G., Zambotti, G., & Lavash, J. (2013b). *Roman road network (version 2008)*. DARMC Scholarly Data Series. Data Contribution Series No. 2013-5.
- Menghin, W. (1995). *Kelten, Römer und Germanen. Archäologie und Geschichte in Deutschland*. Augsburg: Weltbild Verlag.
- Michaels, G., & Rauch, F. (2017). Resetting the urban network: 117–2012. *Economic Journal*. doi:[10.1111/econj.12424](https://doi.org/10.1111/econj.12424).
- Michalopoulos, S., & Papaioannou, E. (2014). National institutions and subnational development in Africa. *Quarterly Journal of Economics*, 129, 151–213.
- Nunn, N. (2008). The long-term effects of Africa's slave trades. *Quarterly Journal of Economics*, 123, 139–176.
- Nüssli, C., & Nüssli, M.-A., (Eds.). (2008). *Georeferenced historical vector data 200*. Euratlas, www.euratlas.com.
- Ochsner, C., & Roesel, F. (2016). *Migrating extremists*. CESifo working paper no. 5799.
- Okoye, D., Pongou, R., & Yokossi, T. (2016). *On the dispensability of new transportation technologies. Evidence from colonial railroads in Nigeria*. MPRA paper no. 75262.
- Pinkovskiy, M. L. (2017). Growth discontinuities at borders. *Journal of Economic Growth*, 22, 145–192.
- Pirenne, H. (1944). *Medieval cities: Their origins and the revival of trade*. Princeton, NJ: Princeton University Press.
- Planck, D., & Beck, W. (1987). *Der Limes in Südwestdeutschland*. Stuttgart: Konrad Theiss Verlag.
- Planitz, H. (1966). *Die deutsche Stadt im Mittelalter. Von der Römerzeit bis zu den Zunftkämpfen*. Wiesbaden: VMA-Verlag.
- Putterman, L. (2008). Agriculture, diffusion and development: Ripple effects of the neolithic revolution. *Economica*, 75, 729–748.
- Rieckhoff, S. (2008). *Geschichte der Chronologie der Späten Eisenzeit in Mitteleuropa und das Paradigma der Kontinuität*. Leipziger Online-Beiträge zur Ur- und Frühgeschichtlichen Archäologie, No. 30.
- Rieckhoff, S., & Biel, J. (Eds.). (2001). *Die Kelten in Deutschland*. Stuttgart: Konrad Theiss Verlag.
- Riemer, U. (2006). *Die Römische Germanienpolitik. Von Caesar bis Commodus*. Darmstadt: Wissenschaftliche Buchgesellschaft.
- Scarre, C. (Ed.). (2000). *Atlas de rome antique*. Paris: Autrement.
- Schallmayer, E. (2011). *Der Limes. Geschichte einer Grenze*. Munich: C.H. Beck.
- Schich, M., Chaoming, S., Yong-Yeol, A., Mirsky, A., Mauro, M., Barabasi, A.-L., et al. (2014). A network framework for cultural history. *Science*, 345, 558–562.
- Schumann, A. (2014). Persistence of population shocks: Evidence from the occupation of West Germany after World War II. *American Economic Journal. Applied Economics*, 6, 189–205.
- Seidel, T., & Von Ehrlich, M. (2015). *The persistent effects of regional policy. Evidence from the West-German Zonenrandgebiet*. CESifo working paper series no. 5373.
- Talbert, R. J. A. (Ed.). (2000). *Barrington atlas of the Greek and Roman world*. Princeton, NJ: Princeton University Press.
- Van Zanden, J. L. (2008). The Road to the Industrial Revolution: Hypotheses and Conjectures about the Medieval Origins of the European Miracle. *Journal of Global History*, 3, 337–359.
- Von Schnurbein, S. (1995). *Vom Einfluss Roms auf die Germanen* (Vol. 331). Vorträge: Nordrhein-westfälische Akademie der Wissenschaften.
- Wolters, R. (2011). *Die Römer in Germanien*. Munich: C.H. Beck.

- Wysokinska, A. (2015). Invisible wall. Role of institutions and culture in long term development. Natural experiment. In *Proceedings of the EEA annual congress 2015*.
- Zabel, F., Putzenlechner, B., & Mauser, W. (2014). Global agricultural land resources. A high-resolution suitability evaluation and its perspectives until 2100 under climate change conditions. *PLoS ONE*, 9, 1–12.